Thermal Stability of Nuclear Reactors

Coupled Neutronic–Thermal Dynamics, Stability Theory, and Historical Lessons from the Chernobyl Accident

A Comprehensive Technical Monograph

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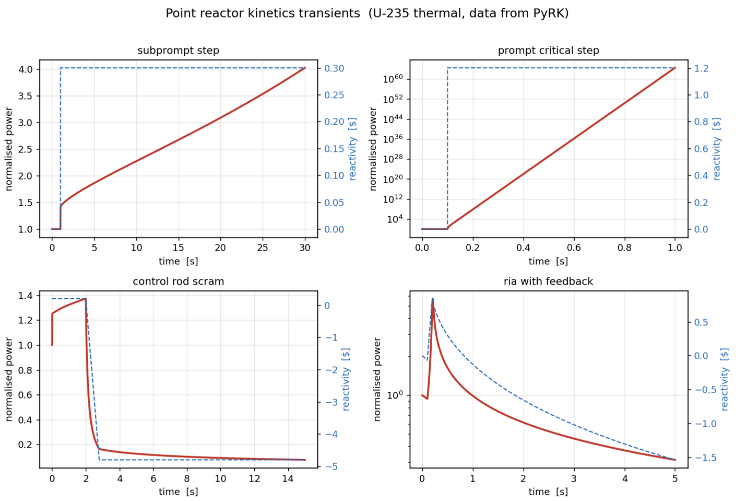

Representative point-reactor-kinetics transients computed with the companion `reactor-kinetics` toolkit.

This document is an educational and historical treatment of reactor dynamics and safety. It is not a design or operating manual.

June 10, 2026

In memory of those who died at Chernobyl, and of the engineers whose hard lessons made every reactor since a little safer.

"Apparently, there was a widespread view that the conditions under which the positive scram effect would be important would never occur. However, they did appear in almost every detail in the course of the actions leading to the Chernobyl accident."

— IAEA, INSAG-7 (1992)

Abstract

Thermal stability is the property that allows a nuclear reactor to respond to perturbations in power, temperature, and reactivity without diverging into an uncontrolled excursion. It is the single most important dynamic characteristic distinguishing a reactor that is safe by its very physics from one that depends on engineered intervention to survive a transient. This monograph develops the theory of reactor thermal stability from first principles and then uses it to dissect the most consequential reactor accident in history—the destruction of Chernobyl Unit 4 on 26 April 1986.

The treatment begins with the neutron balance and the point reactor kinetics equations, including the central role of delayed neutrons, the inhour equation, and the prompt-jump approximation. It then introduces the thermal source term, decay heat, and the conduction–convection chain that carries fission energy from fuel to coolant. With this groundwork, the book defines the reactivity feedback coefficients—Doppler (fuel temperature), moderator and coolant temperature, and the void coefficient—and shows how their signs and magnitudes determine whether the coupled neutronic–thermal-hydraulic system is self-correcting or self-amplifying.

A dedicated treatment of stability theory follows: linearisation of the coupled equations, transfer functions and the Nyquist criterion, the nonlinear Nordheim–Fuchs model of a self-limiting power excursion, and xenon-induced spatial oscillations. The historical part surveys earlier stability-related events (SL-1, Fermi-1, BWR oscillation events) before turning to the RBMK-1000 reactor, whose positive void coefficient and flawed control-rod design set the stage for Chernobyl. The accident is reconstructed step by step and then analysed through the lens of the feedback theory developed earlier, including the evolution of the official understanding from INSAG-1 (1986) to INSAG-7 (1992). The work closes with the principles of inherent and passive safety in modern reactor design and a computational demonstration using a companion open-source point-kinetics toolkit.

Throughout, the emphasis is on the physical meaning of the mathematics: *a reactor is thermally stable when rising temperature removes reactivity faster than rising power can add it.* Chernobyl is, in this language, the story of a machine in which that inequality was reversed.

Preface and How to Read This Book

This book is written for advanced undergraduates, graduate students, and practising engineers who want a unified, self-contained account of why reactors are (or are not) stable, and what happens physically when stability is lost. It assumes a working knowledge of multivariable calculus, ordinary differential equations, and undergraduate physics. No prior course in nuclear engineering is strictly required: the necessary reactor physics is built up from the neutron balance in Chapters 2 and 3.

Organisation

The book is organised into five parts and a set of appendices.

Part I (Foundations of Reactor Dynamics) establishes the language: fission, the multiplication factor and reactivity, the point reactor kinetics equations and the indispensable role of delayed neutrons, and the decay-heat source that persists long after the chain reaction stops.

Part II (Heat Transport and Reactivity Feedback) follows the energy from its birth in the fuel to its removal by the coolant, and then introduces the feedback coefficients that close the loop between temperature and reactivity.

Part III (Stability Theory) treats stability quantitatively, first by linearisation and frequency-domain methods, then through the nonlinear physics of power excursions, and finally through the slow, spatial instabilities driven by xenon-135.

Part IV (Historical Lessons) applies the theory to real machines. After a survey of earlier events, it examines the RBMK-1000 design in detail, then reconstructs and analyses the Chernobyl accident.

Part V (Modern Practice and Synthesis) describes how the lessons are embodied in contemporary designs that aim for inherent and passive safety, and demonstrates the dynamics computationally before drawing the threads together.

A note on conventions

Reactivity is expressed both as an absolute fraction ρ (units of $\Delta k/k$, often quoted in per-cent-mille, pcm, where $1\,\text{pcm} = 10^{-5}$) and in *dollars*, ρ/β, where β is the total delayed-neutron fraction. One dollar of reactivity is the boundary of prompt criticality and is the natural yardstick for transient severity. Temperatures are in kelvin or degrees Celsius as convenient. SI units are used throughout. Numerical examples use validated delayed-neutron data for ^{235}U; the full data set appears in Appendix A.

Companion software

Several figures and worked transients in this book were produced with a small, open-source point-reactor-kinetics toolkit written by the author. The code, which couples validated kinetics data to a stiff ODE solver with temperature feedback, is referenced in Chapter 16 and is freely available. Readers are encouraged to reproduce and extend the examples.

Scope and disclaimer

This is an educational and historical work. It deliberately discusses reactor behaviour at the level of physical principle and published, openly available engineering knowledge. It is *not* an operating procedure, a design specification, or a safety-analysis report, and nothing in it should be used as such.

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Nomenclature

The following symbols are used throughout. Where a symbol has a conventional secondary meaning, the intended sense is clear from context.

Symbol	Meaning
k_{eff}	Effective neutron multiplication factor
ρ	Reactivity, $\rho = (k_{\mathrm{eff}} - 1)/k_{\mathrm{eff}}$
$\beta,\ \beta_i$	Total and group-i delayed-neutron fraction
Λ	Prompt neutron generation time
ℓ	Prompt neutron lifetime
λ_i	Decay constant of delayed-neutron precursor group i
$\zeta_i,\ C_i$	Delayed-neutron precursor concentration (group i)
$p,\ P$	Normalised power; thermal power [W]
n	Neutron density / population
s	Complex frequency (Laplace variable); inverse period $1/T$
T	Reactor period [s]
$T_f,\ T_c$	Fuel and coolant temperature [K]
α_T	Temperature coefficient of reactivity [Δk/K]
α_D	Doppler (fuel-temperature) coefficient
α_v	Void coefficient of reactivity
α_P	Power coefficient of reactivity
$C,\ c_p$	Heat capacity [J/K]; specific heat [J/(kg·K)]
h	Heat transfer coefficient [W/(m^2·K)]
k_f	Fuel thermal conductivity [W/(m·K)]
$\sigma,\ \Sigma$	Microscopic / macroscopic cross-section
ϕ	Neutron flux [n/(cm^2·s)]
$I,\ X$	Iodine-135 and Xenon-135 concentrations
γ_I, γ_X	Fission yields of ^{135}I and ^{135}Xe
\$	"Dollar" of reactivity, ρ/β
pcm	per-cent-mille, $10^{-5}\ \Delta k/k$

Selected abbreviations. LWR (light-water reactor), PWR (pressurised-water reactor), BWR (boiling-water reactor), RBMK (*reaktor bolshoy moshchnosti kanalnyy*, high-power channel-type reactor), MTC (moderator temperature coefficient), FTC (fuel temperature coefficient), ORM (operating reactivity margin), LOCA (loss-of-coolant accident), RIA (reactivity-initiated accident), ECCS (emergency core cooling system), SCRAM (rapid shutdown), INSAG (International Nuclear Safety Advisory Group).

Part I

Foundations of Reactor Dynamics

CHAPTER 1

Introduction: What Stability Means for a Reactor

1.1 The central question

A nuclear reactor is a device that sustains a controlled fission chain reaction and converts the released energy into heat. The adjective *controlled* carries the entire weight of reactor engineering. An uncontrolled chain reaction is not, in itself, exotic: it is the natural tendency of an assembly of fissile material that is more than critical. What makes a reactor useful—and safe—is that it can be held at a steady, prescribed power, and that when it is disturbed it returns toward a steady state rather than running away.

This property is **thermal stability**. Informally, a reactor is thermally stable if an increase in power produces, through the heating of its materials, a change in conditions that tends to *reduce* power again. The feedback loop closes through temperature: power heats the fuel and coolant, and the resulting temperature changes alter the neutron economy. If that alteration opposes the original power change, the reactor is self-regulating; if it reinforces the change, the reactor is unstable, and a small perturbation can grow without bound until something physical—melting, vaporisation, mechanical disassembly—intervenes.

> **Key point.** A reactor is thermally stable when a rise in temperature removes reactivity faster than the rise in power can add it. Stability is therefore a statement about the *sign* and *magnitude* of the reactivity feedback coefficients, weighed against the speed of the underlying neutron kinetics.

The purpose of this book is to make that informal statement precise, to develop the mathematics that quantifies it, and to show—through the Chernobyl accident—what happens when the inequality is violated by design and by operation simultaneously.

1.2 Why stability is non-trivial

It is tempting to think that stability is automatic: surely any sensible machine settles down after a nudge. Three features of reactor physics make the question genuinely subtle.

First, the kinetics are fast. The prompt neutron generation time Λ in a thermal reactor is of order 10^{-5} s to 10^{-4} s. If the chain reaction depended only on prompt neutrons, a reactivity insertion of even a fraction of a per cent would double the power in milliseconds—far faster than any mechanical control could respond, and far faster than heat could redistribute to provide feedback. The reactor would be, for practical purposes, uncontrollable.

Second, delayed neutrons rescue controllability. A small fraction $\beta \approx 0.0065$ for ^{235}U) of fission neutrons are emitted not promptly but seconds to minutes later, from the decay of specific fission products. These *delayed neutrons* slow the effective response of the reactor by orders of magnitude, stretching the characteristic timescale from microseconds to seconds when the reactivity is kept below β. The entire practice of reactor control lives in the narrow margin below one dollar of reactivity, where delayed neutrons dominate the transient. Cross that boundary—become *prompt-critical*—and the delayed neutrons no longer matter; the reactor reverts to its microsecond timescale. Chernobyl crossed it.

Third, feedback is both stabilising and slow, or fast and destabilising, depending on design. The Doppler effect in the fuel acts almost instantaneously because it depends on the fuel temperature, which responds immediately to a power change. Moderator and coolant feedback are slower because heat must first cross the fuel–clad–coolant path. Void feedback—the effect of steam bubbles in the coolant—can be fast and, crucially, can have either sign. Whether the *net* of these effects stabilises the reactor depends on the reactor type, the fuel loading, the coolant state, and even the point in the fuel cycle. There is nothing automatic about it.

1.3 A first picture of the feedback loop

Figure 1.1 shows the structure we will formalise. External reactivity (control rods, boron, operator actions) drives the neutron kinetics, which set the power. The power deposits heat, which raises temperatures and changes the coolant state. Those thermal changes feed back as additional reactivity. The closed loop can be self-correcting (negative feedback) or self-amplifying (positive feedback).

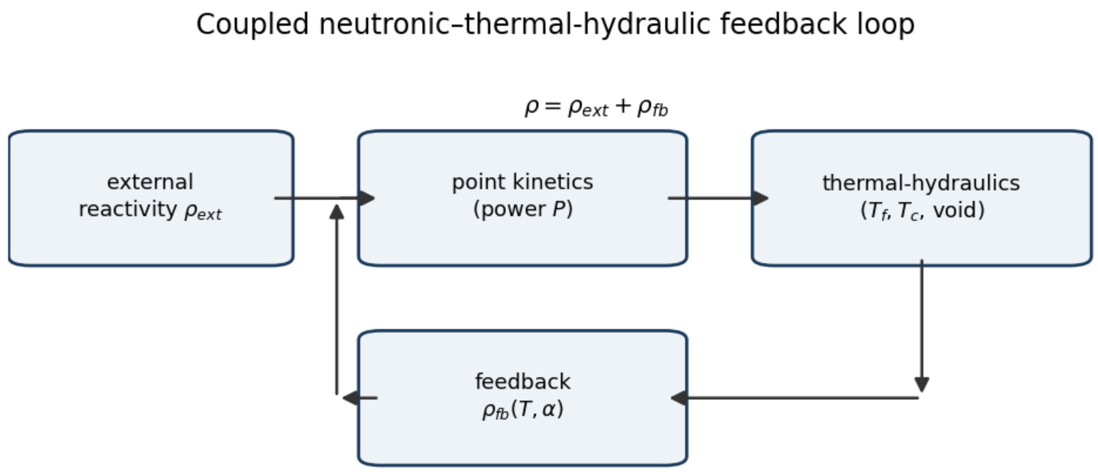

Figure 1.1: The coupled neutronic–thermal-hydraulic feedback loop. The sign of the feedback path $\rho_{fb}(T, \alpha)$ determines stability.

The mathematics of the upper path (kinetics) occupies Chapter 3; the right-hand block (thermal-hydraulics) occupies Chapter 5; the feedback path occupies Chapter 6; and the behaviour of the *closed* loop—its stability—occupies Part III.

1.4 Two reactors, two fates

To fix ideas, contrast two reactor philosophies that recur throughout the book.

A **pressurised-water reactor (PWR)** is deliberately *under-moderated*. Its lattice is designed so that if the water heats and expands, or if it boils and forms voids, the moderation of neutrons *worsens*, the chain reaction weakens, and power falls. Both its moderator temperature coefficient and its void coefficient are negative. Combined with a strongly negative Doppler coefficient in the fuel, the PWR is inherently self-regulating across its operating range: pushing the power up pushes the temperature up, which pushes the reactivity—and hence the power—back down.

The **RBMK-1000**, the Soviet design at Chernobyl, was *over-moderated* by virtue of its large graphite moderator, which is independent of the water coolant. In the RBMK, the water served mainly as coolant and as a neutron *absorber*. Boiling the water therefore *removed an absorber* without removing much moderation, so steam voids *increased* reactivity. Its void coefficient was strongly positive, especially at low power and with the core configured as it was on the night of the accident. Pushing the power up boiled more water, which pushed the reactivity—and the power—further up. The same nudge that a PWR damps, the RBMK amplified.

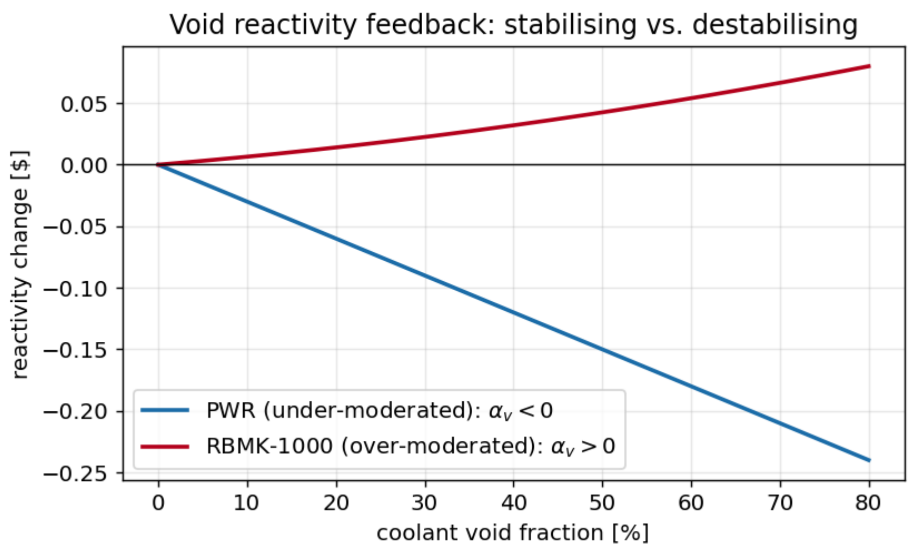

Figure 1.2: Schematic void reactivity feedback. In an under-moderated lattice (PWR) voids reduce reactivity; in an over-moderated lattice (RBMK) voids increase it. The slope at the operating point is the void coefficient α_v.

This single contrast—under-moderated versus over-moderated, negative versus positive void coefficient—is the physical heart of the Chernobyl story, and we will return to it repeatedly with increasing precision.

1.5 What "thermal" stability is not

It is worth separating thermal stability from several neighbouring concepts that are sometimes conflated with it.

Criticality is a static balance: $k_{\text{eff}} = 1$, neutron production equals neutron loss. A reactor can be exactly critical and still be dynamically unstable. Stability is about what happens to small departures from balance, not about the balance itself.

Controllability is the ability of operators and automatic systems to drive the reactor to a desired state. A reactor can be controllable yet rely on that control to remain safe; the ideal is a reactor whose physics keep it safe even when control fails. Inherent stability reduces the burden on control.

Thermal-hydraulic stability in the narrow sense refers to flow and boiling instabilities (density-wave oscillations, flow excursions) that can occur even at fixed power. These couple strongly to neutronics in boiling systems and are part of the broader stability picture treated in Chapter 8.

Structural and material stability—the integrity of fuel, cladding, and pressure boundary under thermal stress—is the ultimate limit that a power excursion challenges. Thermal stability theory predicts whether and how fast a reactor approaches those limits.

1.6 Roadmap and how the Chernobyl thread runs through the book

Each part of this book develops a piece of theory and then points it at Chernobyl. Part I gives us the kinetics that explain why the final excursion was a prompt-critical event measured in seconds. Part II gives us the feedback coefficients that explain why the RBMK amplified rather than damped the disturbance. Part III gives us the stability criteria that, applied to the RBMK's operating state that night, predict instability. Part IV reconstructs the accident and shows the theory accounting for it in detail. Part V asks how modern designs build in the negative feedback that the RBMK lacked, and demonstrates the dynamics with a small open-source kinetics code.

The recurring lesson is not that the operators were uniquely reckless or that one component uniquely failed. It is that a reactor with the wrong sign of feedback, operated into the corner of its envelope where that wrong sign was strongest, behaved exactly as the equations say such a machine must. Stability is physics; the accident was the physics asserting itself.

Exercises

1. Explain in one sentence the difference between criticality and stability, and give an example of a reactor state that is critical but unstable.
2. A reactor has a positive void coefficient. Describe qualitatively the sequence of events following a small increase in power, and contrast it with a reactor having a negative void coefficient.
3. Why is it insufficient for a reactor merely to "settle down" after a disturbance for it to be called inherently safe? Distinguish stability from controllability.
4. List the three features of reactor physics that make stability non-trivial, and explain which one delayed neutrons mitigate.
5. Using the under-/over-moderated distinction, explain why a PWR and an RBMK respond oppositely to coolant boiling.

CHAPTER 2

Reactor Physics Fundamentals

This chapter assembles the minimum reactor physics needed for a dynamic theory: fission and the neutron balance, the multiplication factor and reactivity, the six-factor formula, and the meaning of the neutron lifetime. Readers with a background in nuclear engineering may skim it; the notation introduced here is used everywhere later.

2.1 Fission and the neutron economy

When a ^{235}U nucleus absorbs a thermal neutron it may fission, splitting into two fragments and releasing on average $\nu \approx 2.43$ prompt neutrons together with about 200 MeV of energy. The energy is distributed roughly as 168 MeV in the kinetic energy of the fragments (deposited essentially at the point of fission, in the fuel), with the remainder in prompt neutrons, prompt gamma rays, and—importantly for later—the radioactive decay of the fission products. Of the recoverable energy, about 93% appears promptly and about 7% is released later as decay heat (Chapter 4).

A self-sustaining chain reaction requires that, on average, exactly one of the ν neutrons from each fission goes on to cause another fission. The other neutrons are lost to non-fission capture (in fuel, moderator, coolant, structure, and control materials) or to leakage out of the core. The bookkeeping of these gains and losses is the neutron economy.

2.2 The multiplication factor and reactivity

Define the effective multiplication factor

$$k_{\text{eff}} = \frac{\text{neutrons produced in one generation}}{\text{neutrons lost (absorbed + leaked) in the preceding generation}}. \tag{2.1}$$

The three regimes are

$$k_{\text{eff}} < 1 \text{ (subcritical)}, \qquad k_{\text{eff}} = 1 \text{ (critical)}, \qquad k_{\text{eff}} > 1 \text{ (supercritical)}. \tag{2.2}$$

A critical reactor sustains a constant population; a supercritical one grows.

Because the interesting dynamics involve small departures from criticality, it is convenient to measure the distance from $k_{\text{eff}} = 1$ directly. The **reactivity** is

$$\boxed{\rho \equiv \frac{k_{\text{eff}} - 1}{k_{\text{eff}}}} \tag{2.3}$$

which is dimensionless, zero at criticality, positive above it, and negative below. Reactivity is small in practice: a useful operating insertion might be a few hundred pcm (recall 1 pcm $= 10^{-5}$), and one "dollar" equals $\beta \approx 650$ pcm for ^{235}U. We will almost always work to first order in ρ, where $\rho \approx k_{\text{eff}} - 1$.

Definition 2.1 (Reactivity in dollars)**.** The reactivity expressed in units of the delayed-neutron fraction, $\$ \equiv \rho/\beta$, is the natural measure of transient severity. At $\rho = \beta$ ($\$ = 1$) the reactor is prompt-critical: the prompt neutrons alone sustain the chain, and delayed neutrons are no longer needed to maintain growth.

2.3 The six-factor formula

The effective multiplication factor for a finite thermal reactor factors into six physically distinct processes:

$$k_{\text{eff}} = \eta\, f\, p\, \varepsilon\, P_f\, P_t, \tag{2.4}$$

where

- η is the reproduction factor (neutrons produced per thermal neutron absorbed in fuel);
- f is the thermal utilisation (fraction of thermal absorptions occurring in fuel rather than in moderator, coolant, or structure);
- p is the resonance escape probability (fraction of neutrons that slow past the ^{238}U capture resonances without being absorbed);
- ε is the fast-fission factor (a small boost from fissions caused by fast neutrons);
- P_f and P_t are the fast and thermal non-leakage probabilities.

The product $\eta f p \varepsilon$ is the infinite-medium multiplication factor k_∞, and $k_{\text{eff}} = k_\infty P_f P_t$.

The value of the six-factor formula for our purposes is that it localises feedback. Temperature and voids do not change k_{eff} abstractly; they change *specific factors*. The Doppler effect (Chapter 6) acts on the resonance escape probability p through the broadening of the ^{238}U resonances. Moderator heating and voiding act on f and p through changes in moderator density and the moderation ratio. Expansion changes leakage, $P_f P_t$. Tracking *which* factor a given feedback alters, and with *which sign*, is the discipline that distinguishes a stable lattice from an unstable one.

> **Key point.** Whether a reactor is under- or over-moderated is, in six-factor terms, a competition: voiding the coolant tends to raise p (less parasitic capture and resonance absorption) but lower f and worsen moderation. In an under-moderated lattice the harmful effects win and reactivity falls; in an over-moderated lattice (like the RBMK with its graphite) the helpful effect on p and the loss of the water absorber win, and reactivity rises.

2.4 The neutron lifetime and generation time

Two closely related timescales govern how fast the population responds.

The **prompt neutron lifetime** ℓ is the mean time from a neutron's birth to its absorption or leakage. In a thermal reactor it is dominated by the slowing-down time plus the thermal diffusion time and is of order 10^{-4} s in a light-water reactor, somewhat longer in a graphite system.

The **prompt neutron generation time** is

$$\Lambda = \frac{\ell}{k_{\text{eff}}} \approx \ell \quad (k_{\text{eff}} \approx 1), \tag{2.5}$$

the mean time between successive neutron generations. We use Λ in the kinetics equations; for ^{235}U thermal systems a representative value is $\Lambda \sim 10^{-5}-10^{-4}$ s. Its smallness is precisely why prompt supercriticality is so dangerous: with $\Lambda = 10^{-5}$ s and a prompt reactivity of just 0.2%, the prompt growth rate is $\rho_{\text{prompt}}/\Lambda \sim 200\,\text{s}^{-1}$, an e-folding every 5 ms.

2.5 Neutron flux, cross-sections, and reaction rates

The neutron flux $\phi(\mathbf{r}, E, t)$ is the product of neutron density and speed; it measures the rate at which neutrons traverse unit area. Reaction rates are products of flux and macroscopic cross-section,

$$R_x = \Sigma_x \phi, \qquad \Sigma_x = N\sigma_x, \tag{2.6}$$

where N is the number density of nuclei and σ_x the microscopic cross-section for reaction x (fission, capture, scattering). Cross-sections depend strongly on neutron energy; the resonance structure of ^{238}U capture near a few eV, and the $1/v$ behaviour of many thermal absorbers, are central to feedback and to xenon poisoning respectively.

The fission power density is

$$q'''(\mathbf{r}, t) = E_f\, \Sigma_f\, \phi(\mathbf{r}, t), \tag{2.7}$$

with $E_f \approx 200\,\text{MeV}$ the recoverable energy per fission. Integrated over the core, this is the thermal power that the kinetics equations track and that the thermal-hydraulic system must remove.

2.6 From the transport equation to point kinetics

The full description of the neutron field is the time-dependent neutron transport (or, in diffusion approximation, the neutron diffusion) equation, a seven-dimensional integro-differential equation in space, energy, angle, and time. For stability analysis we usually do not need its full spatial detail. If the spatial shape of the flux changes slowly compared with its amplitude—the *point* approximation—we can separate variables,

$$\phi(\mathbf{r}, E, t) \approx \psi(\mathbf{r}, E)\, n(t), \tag{2.8}$$

and reduce the dynamics to ordinary differential equations for the amplitude $n(t)$ and the delayed-neutron precursors. This reduction yields the point reactor kinetics equations of Chapter 3. The approximation is excellent for compact cores responding coherently, and it is the workhorse of reactor dynamics. Its principal failure—when different regions of a large core behave differently—gives rise to the spatial instabilities of Chapter 10, and was a real complicating factor in the large RBMK core at Chernobyl, where the point picture understates the danger of an asymmetric, axially-peaked flux.

2.7 One-group diffusion theory and the criticality eigenvalue

The point-kinetics amplitude equation rests on separating the neutron field into a fixed spatial shape and a time-dependent amplitude. We justify this from one-group diffusion theory and, in the process, define criticality as an eigenvalue problem.

In the one-group diffusion approximation the flux $\phi(\mathbf{r}, t)$ obeys

$$\frac{1}{v}\frac{\partial \phi}{\partial t} = D\nabla^2\phi - \Sigma_a\phi + \nu\Sigma_f\phi \quad \text{in } \Omega, \qquad \phi|_{\partial\Omega} = 0, \tag{2.9}$$

with diffusion coefficient D, absorption Σ_a and production $\nu\Sigma_f$. Seeking separable solutions $\phi = \psi(\mathbf{r})T(t)$ forces the spatial factor to satisfy the Helmholtz eigenproblem

$$\nabla^2\psi + B^2\psi = 0 \text{ in } \Omega, \qquad \psi|_{\partial\Omega} = 0, \tag{2.10}$$

where B^2 is the *geometric buckling*: the spectrum $0 < B_0^2 < B_1^2 \leq \cdots$ consists of the Dirichlet eigenvalues of $-\nabla^2$, and the fundamental eigenfunction $\psi_0 > 0$ is the unique sign-definite mode.

Theorem 2.2 (Criticality condition)**.** *With $\phi = \psi_0(\mathbf{r})e^{\omega t}$ inserted into* (2.9),

$$\frac{\omega}{v} = -DB_0^2 - \Sigma_a + \nu\Sigma_f. \tag{2.11}$$

Defining $L^2 = D/\Sigma_a$ and $k_\infty = \nu\Sigma_f/\Sigma_a$, the reactor is critical ($\omega = 0$) *iff*

$$k_{\text{eff}} \equiv \frac{k_\infty}{1 + L^2B_0^2} = 1. \tag{2.12}$$

Proof. Insert $\phi = \psi_0 e^{\omega t}$ into (2.9) and use $\nabla^2\psi_0 = -B_0^2\psi_0$; dividing by $\psi_0 e^{\omega t}$ gives the dispersion relation. Setting $\omega = 0$ yields $\nu\Sigma_f = \Sigma_a + DB_0^2$; dividing by $\Sigma_a(1 + L^2B_0^2)$ and identifying k_∞, L^2 gives (2.12). □

Equation (2.12) is the diffusion-theory form of the six-factor formula: the factor $(1 + L^2B_0^2)^{-1}$ is exactly the non-leakage probability, and B_0^2 carries the geometry (e.g. $B_0^2 = (\pi/\tilde{R})^2$ for a bare slab or sphere of extrapolated half-width/radius $\tilde{R}$). The dominance of the fundamental mode is what licenses the point-kinetics ansatz.

Proposition 2.3 (Fundamental-mode dominance)**.** *Expanding $\phi = \sum_n a_n(t)\psi_n$ in Helmholtz eigenfunctions, each amplitude obeys $\dot{a}_n/v = (\nu\Sigma_f - \Sigma_a - DB_n^2)a_n$, so $a_n(t) = a_n(0)e^{\omega_n t}$ with ω_n strictly decreasing in B_n^2. Hence $\omega_0 > \omega_n$ for all $n \geq 1$, $a_n/a_0 \to 0$ as $t \to \infty$, and $\phi(\mathbf{r}, t) \to \psi_0(\mathbf{r})\, n(t)$. The point-kinetics separation is the asymptotic, fundamental-mode limit.*

Proof. Orthogonality of the ψ_n under (2.10) decouples (2.9) mode by mode, giving the stated scalar ODEs; the ordering $\omega_0 > \omega_1 > \cdots$ follows from $B_0^2 < B_1^2 < \cdots$, and the ratios $a_n(t)/a_0(t) = [a_n(0)/a_0(0)]\, e^{(\omega_n - \omega_0)t} \to 0$. □

Exercises

1. Starting from $k_{\text{eff}} = 1.002$, compute the reactivity ρ in pcm and in dollars (take $\beta = 0.0065$).

2. Using the six-factor formula, identify which factor the Doppler effect modifies and with what sign.
3. Estimate the prompt growth rate $\rho_{\text{prompt}}/\Lambda$ for $\rho_{\text{prompt}} = 0.003$ and $\Lambda = 10^{-5}\,\text{s}$, and the corresponding e-folding time.
4. Explain physically why voiding an over-moderated lattice raises the resonance escape probability p.
5. Why does the point-kinetics approximation fail for large cores, and what phenomenon (Chapter 10) results?

CHAPTER 3

Point Reactor Kinetics

The point reactor kinetics (PRK) equations are the dynamical backbone of this book. They describe how reactor power evolves in time under a prescribed reactivity, and they make explicit the decisive role of delayed neutrons. We derive them, solve them in the cases that matter for stability, and extract the two results—the inhour equation and the prompt-jump approximation—that we will use again and again.

3.1 Delayed neutrons

Most fission neutrons are emitted within 10^{-14} s of fission: these are *prompt*. A small fraction, however, appear with a delay, emitted by certain neutron-rich fission products ("precursors") as they beta-decay. The classic example is ^{87}Br, which decays to an excited ^{87}Kr that promptly emits a neutron; the effective delay is set by the 55 s half-life of the bromine. In practice the many precursors are grouped into six (sometimes eight) effective groups, each with a yield fraction β_i and a decay constant λ_i.

The total delayed fraction is

$$\beta = \sum_{i=1}^{6} \beta_i \approx 0.0065 \text{ for } {}^{235}\text{U (thermal)}, \tag{3.1}$$

with group decay constants ranging from $\lambda_1 \approx 0.0124\,\text{s}^{-1}$ (half-life ~ 56 s) to $\lambda_6 \approx 3.0-3.9\,\text{s}^{-1}$ (half-life under a second). Although less than one neutron in a hundred is delayed, these neutrons set the timescale of the controllable reactor: their mean delay, of order 10 s, vastly exceeds the prompt generation time $\Lambda \sim 10^{-5}$ s, so the *average* neutron lifetime weighted by prompt and delayed contributions is dominated by the delayed term whenever the reactor is below prompt criticality. Appendix A lists the full data set.

3.2 Derivation of the PRK equations

Let $n(t)$ be the neutron population (proportional to power P) and $C_i(t)$ the concentration of precursor group i. In one generation, k_{eff} neutrons are produced per neutron; of these, a fraction $(1-\beta)$ are prompt and a fraction β_i feed precursor group i. The prompt balance gives a production rate $(1-\beta)k_{\text{eff}}n/\ell$ against a loss rate n/ℓ, while each precursor group gains at rate $\beta_i k_{\text{eff}} n/\ell$ and loses by decay at rate $\lambda_i C_i$; the decay of precursors returns neutrons at rate $\lambda_i C_i$. Writing $\Lambda = \ell/k_{\text{eff}}$ and using $\rho = (k_{\text{eff}}-1)/k_{\text{eff}}$, the result is the standard six-group point reactor kinetics equations:

$$\boxed{\begin{aligned} \frac{\mathrm{d}n}{\mathrm{d}t} &= \frac{\rho(t)-\beta}{\Lambda}\,n(t) + \sum_{i=1}^{6}\lambda_i C_i(t), \\ \frac{\mathrm{d}C_i}{\mathrm{d}t} &= \frac{\beta_i}{\Lambda}\,n(t) - \lambda_i C_i(t), \qquad i = 1,\dots,6. \end{aligned}} \tag{3.2}$$

The same equations hold with n replaced by power P, since the two are proportional in the point approximation. In normalised form we write $p = P/P_0$.

The structure of (3.2) is already revealing. The prompt coefficient $(\rho-\beta)/\Lambda$ changes sign at $\rho = \beta$. Below it, the coefficient is *negative*: the prompt chain alone would die away, and the population is held up only by the slow return of delayed neutrons through $\sum \lambda_i C_i$. This is the regime of controllable operation. Above it, $\rho > \beta$, the prompt coefficient is positive and the population grows on the Λ timescale regardless of the delayed neutrons—prompt criticality.

3.3 Steady state and the initial condition

At constant power p_0 with $\rho = 0$, the time derivatives vanish and

$$C_{i,0} = \frac{\beta_i}{\lambda_i \Lambda}\,p_0, \tag{3.3}$$

which is the equilibrium precursor inventory. The precursors hold a large reservoir of "stored" delayed neutrons: because $\lambda_i\Lambda \ll \beta_i$, the equilibrium $C_{i,0}$ greatly exceeds p_0. This reservoir is what gives the reactor its inertia and is the physical reason the prompt-jump (below) is finite.

3.4 The prompt jump approximation

Consider a step insertion of reactivity $\rho < \beta$ at $t = 0$ on a reactor at steady power. Immediately after the step, the precursor concentrations cannot change instantaneously (they evolve on the slow λ_i timescale), so the $\sum \lambda_i C_i$ term is momentarily fixed at its pre-step value. The fast prompt dynamics then drive n to a new quasi-equilibrium on the Λ timescale. Setting $\mathrm{d}n/\mathrm{d}t \to 0$ in (3.2) with the precursor term held fixed yields the **prompt-jump approximation**:

$$\boxed{\frac{n(0^+)}{n(0^-)} = \frac{\beta}{\beta-\rho}} \tag{3.4}$$

The power jumps essentially instantly by this factor, then rises (or falls) more slowly as the precursors readjust. For $\rho = 0.5\$$ ($\rho = 0.5\beta$) the jump is a factor of two; as $\rho \to \beta^-$ the predicted jump diverges, signalling the breakdown of the approximation and the onset of prompt criticality. The prompt-jump captures the "instant" part of a transient and is invaluable for estimating the immediate consequence of a reactivity insertion.

3.5 The inhour equation

To find the asymptotic behaviour, seek exponential solutions $n, C_i \propto e^{st}$ for constant ρ. Substituting into (3.2) gives $C_i = \beta_i n/[\Lambda(s+\lambda_i)]$, and (3.2) becomes the **inhour equation**:

$$\boxed{\rho = \Lambda s + \sum_{i=1}^{6} \frac{\beta_i\, s}{s+\lambda_i}} \tag{3.5}$$

This transcendental equation has seven real roots s_j for six delayed groups. The roots interlace the negative poles $-\lambda_i$; six of them are negative (decaying transients), and one—the *dominant* root—has the sign of ρ. The reactor period is $T = 1/s_{\text{dominant}}$, the time for the power to change by a factor e.

For small positive reactivity the dominant root is small, $s \approx \rho/(\beta-\rho)\cdot\bar{\lambda}^{-1}$ in a one-group caricature, giving a long (stable, slow) period dominated by the delayed neutrons. As $\rho \to \beta$ the dominant root grows rapidly and the period collapses; beyond β the leading behaviour is $s \approx (\rho-\beta)/\Lambda$, the prompt period.

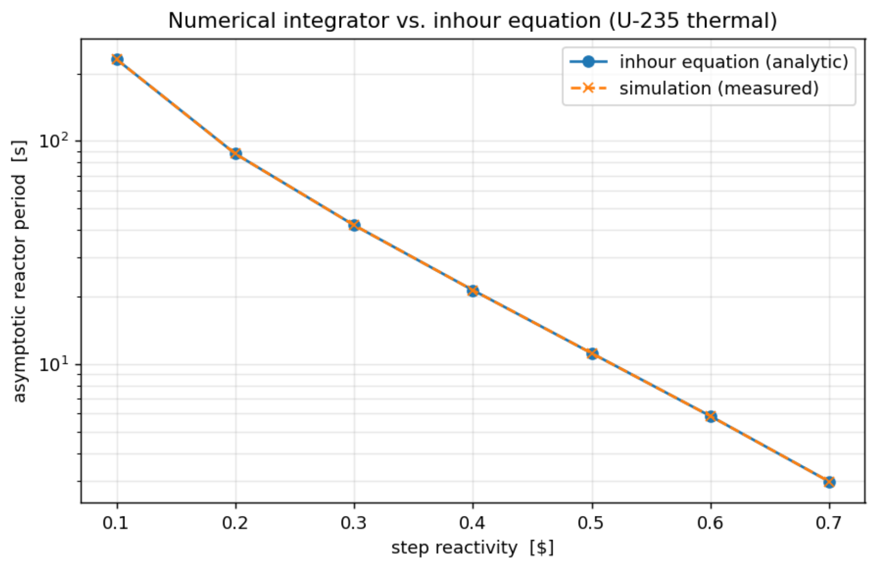

Figure 3.1: Asymptotic reactor period as a function of step reactivity for ^{235}U thermal data. The closed-form inhour solution (solid) and a direct numerical integration of (3.2) (dashed) agree to better than 10^{-4}. Note the steep collapse of the period as reactivity approaches one dollar.

Example 3.1 (The danger of one dollar)**.** With $\Lambda = 10^{-5}\,\text{s}$ and $\beta = 0.0065$, a reactivity of $0.5\$ = 325\,\text{pcm}$ gives a stable period of tens of seconds—easily controllable. A reactivity of $1.5\$ = 975\,\text{pcm}$ is $0.5\$$ above prompt critical; the dominant root is approximately $s \approx (\rho-\beta)/\Lambda = (0.00975-0.0065)/10^{-5} \approx 325\,\text{s}^{-1}$, an e-folding every 3 ms. Power rises by a factor of $e^{10} \approx 2\times 10^4$ in just 30 ms. No mechanical system can intervene on that timescale; only the prompt negative feedback of the fuel (Doppler) can, if it exists and has the right sign.

3.6 Numerical solution and stiffness

For time-varying $\rho(t)$, or with feedback, (3.2) must be integrated numerically. The system is *stiff*: it contains the very fast prompt timescale Λ alongside the slow precursor timescales $1/\lambda_i$, spanning six orders of magnitude. Explicit integrators require prohibitively small steps; implicit or specialised stiff methods (backward differentiation, LSODA, or analytic-within-a-step schemes) are standard. Figure 3.2 shows representative transients computed this way: a sub-prompt step settling onto a stable period, a prompt-critical excursion diverging on the millisecond scale, a control-rod scram, and—most importantly for stability—a reactivity-initiated transient that is turned over by negative temperature feedback. The last of these is the prototype of a *thermally stable* response, and the contrast with the un-fed-back prompt-critical case is the whole subject of the book in one picture.

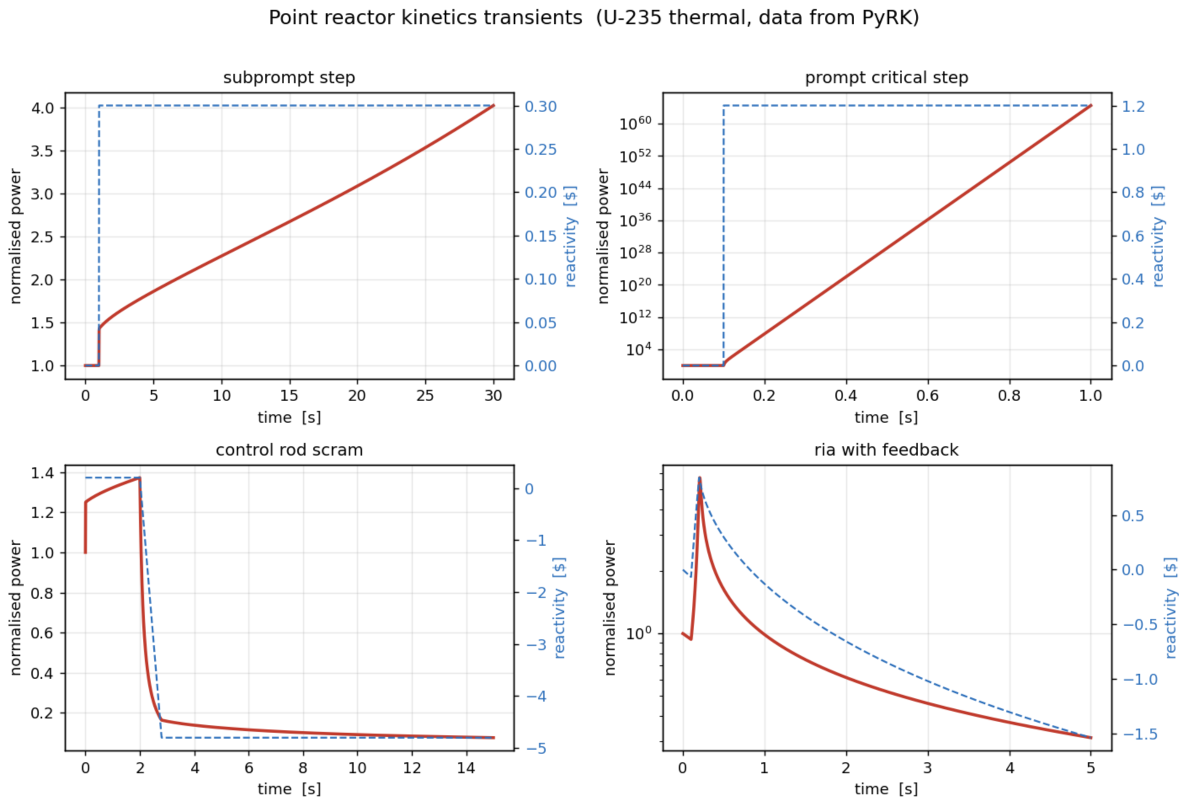

Figure 3.2: Point-kinetics transients for ^{235}U thermal data. Top left: a sub-prompt +0.3$ step (stable period). Top right: a prompt-critical +1.2$ step (runaway, log scale). Bottom left: a control-rod scram. Bottom right: a reactivity-initiated excursion limited by negative feedback—the signature of thermal stability.

Key point. The point kinetics equations contain the entire timescale problem of reactor control. Below one dollar, delayed neutrons stretch the response to seconds and the reactor is controllable. Above one dollar, the response collapses to the prompt timescale and only instantaneous feedback can save the reactor. Everything that follows—feedback coefficients, stability criteria, the Chernobyl excursion—is about staying below that line, or about what happens when a machine crosses it.

3.7 Matrix formulation and spectral theory

Write the state $\mathbf{y} = (n, C_1, \ldots, C_G)^\top \in \mathbb{R}^{G+1}$. For constant ρ the point kinetics (3.2) are the linear system $\dot{\mathbf{y}} = \mathbf{M}(\rho)\mathbf{y}$ with

$$\mathbf{M}(\rho) = \begin{pmatrix} \dfrac{\rho-\beta}{\Lambda} & \lambda_1 & \cdots & \lambda_G \\ \dfrac{\beta_1}{\Lambda} & -\lambda_1 & & \\ \vdots & & \ddots & \\ \dfrac{\beta_G}{\Lambda} & & & -\lambda_G \end{pmatrix}. \tag{3.6}$$

Theorem 3.2 (Spectrum equals inhour roots). *$s \in \sigma(\mathbf{M}(\rho))$ if and only if s solves the inhour equation* (3.5)*: $\rho = \Lambda s + \sum_{i=1}^{G} \beta_i s/(s+\lambda_i)$.*

Proof. For an eigenpair $(s, \mathbf{y})$ the precursor rows give $C_i = \frac{\beta_i}{\Lambda(s+\lambda_i)} n$ (for $s \neq -\lambda_i$), and the first row gives $s\,n = \frac{\rho-\beta}{\Lambda} n + \sum_i \lambda_i C_i$. Substituting and dividing by n/Λ, $\Lambda s = \rho - \beta + \sum_i \frac{\lambda_i \beta_i}{s+\lambda_i}$; using $\lambda_i/(s+\lambda_i) = 1 - s/(s+\lambda_i)$ yields (3.5). The poles $s = -\lambda_i$ are not eigenvalues (they force $n = 0$ and hence $\mathbf{y} = 0$). □

Theorem 3.3 (Reality, simplicity, interlacing). *For $\rho \neq 0$ the inhour equation has exactly $G+1$ real simple roots $s_0 > s_1 > \cdots > s_G$ obeying $s_0 > -\lambda_{(1)} > s_1 > -\lambda_{(2)} > \cdots > -\lambda_{(G)} > s_G$, where $\lambda_{(1)} < \cdots < \lambda_{(G)}$ are the ordered decay constants. Moreover* $\operatorname{sign}(s_0) = \operatorname{sign}(\rho)$.

Proof. Let $h(s) = \Lambda s + \sum_i \beta_i s/(s+\lambda_i)$. Then $h'(s) = \Lambda + \sum_i \beta_i \lambda_i/(s+\lambda_i)^2 > 0$ wherever defined, so h is strictly increasing on each interval between consecutive poles, running from $-\infty$ to $+\infty$ across each; thus $h(s) = \rho$ has exactly one root in each of the G inter-pole gaps and one to the right of $-\lambda_{(1)}$, giving $G+1$ simple real interlacing roots. Since $h(0) = 0$ and h is increasing through the origin, $\rho > 0 \Rightarrow s_0 > 0$ and $\rho < 0 \Rightarrow s_0 < 0$. □

Corollary 3.4. *The asymptotic period $T = 1/s_0$ has the sign of ρ, and the solution is $\mathbf{y}(t) = \sum_{j=0}^{G} c_j \mathbf{v}_j e^{s_j t}$ with $\mathbf{v}_j$ the eigenvector of s_j; for $t \to \infty$ the dominant mode $e^{s_0 t}$ prevails.*

Proposition 3.5 (Nonnegativity of the flow). *$\mathbf{M}(\rho)$ is a Metzler matrix (off-diagonal entries ≥ 0); hence $e^{\mathbf{M}t} \geq 0$ entrywise for $t \geq 0$, and $\mathbf{y}(0) \geq 0$ implies $\mathbf{y}(t) \geq 0$ for all $t \geq 0$. In particular the power $n(t)$ never goes negative.*

Proof. The off-diagonal entries are $\lambda_i \geq 0$ and $\beta_i/\Lambda \geq 0$; all other off-diagonal entries vanish. Choose $c > 0$ with $\mathbf{M} + c\mathbf{I} \geq 0$; then $e^{\mathbf{M}t} = e^{-ct} e^{(\mathbf{M}+c\mathbf{I})t}$, and the exponential of a nonnegative matrix is nonnegative (its power series has nonnegative terms). A nonnegative semigroup maps the nonnegative orthant into itself. □

Prompt jump as a singular perturbation. Treat Λ as a small parameter. The fast equation $\Lambda \dot{n} = (\rho - \beta)n + \Lambda \sum_i \lambda_i C_i$ has, for $\rho < \beta$, a boundary layer of width $\mathcal{O}\big(\Lambda/(\beta-\rho)\big)$ during which the slow variables C_i are frozen at their pre-step values. Matching the inner (layer) solution to the outer quasi-static balance $n = \Lambda\big(\sum_i \lambda_i C_i\big)/(\beta - \rho)$ gives, at $t = 0^+$, $n(0^+)/n(0^-) = \beta/(\beta - \rho)$, i.e. (3.4), with relative error $\mathcal{O}(\Lambda)$. This places the prompt-jump approximation on a rigorous asymptotic footing.

Exercises

1. Derive the steady-state precursor concentration (3.3) from the precursor equation.
2. Compute the prompt-jump factor for $\rho = 0.25\$$ and for $\rho = 0.9\$$. Comment on the trend as $\rho \to \beta$.
3. Show that the inhour equation (3.5) has exactly $G + 1$ real roots for G delayed groups.
4. For $\rho = 1.5\$$, $\Lambda = 10^{-5}\,\mathrm{s}$, $\beta = 0.0065$, estimate the prompt period and the factor by which power grows in $20\,\mathrm{ms}$.
5. Explain why the point-kinetics system is numerically stiff and why explicit integrators struggle with it.

CHAPTER 4

Decay Heat: The Source That Will Not Switch Off

4.1 Why decay heat matters for stability and safety

The point kinetics equations describe the *fission* power, which responds within seconds to a scram. But roughly 7% of a reactor's steady-state thermal power comes not from fission directly but from the radioactive decay of fission products accumulated in the fuel. This **decay heat** cannot be switched off by inserting control rods: it is governed by the half-lives of hundreds of nuclides and decays only with time. Immediately after shutdown from full power, decay heat is about $6-7\%$ of the previous power; it falls to roughly 1% after an hour and continues to decline over days. For a 3000 MW thermal core, 7% is 210 MW—a space-heater the size of a small power station that turns on the instant you shut the reactor down and that must be removed continuously to prevent the fuel from melting.

Decay heat is the reason a reactor is never truly "off" from a thermal standpoint, and it is the link between a reactivity transient and a thermal catastrophe: at Chernobyl, at Three Mile Island, and at Fukushima, decay heat was the persistent source that turned a loss of cooling into core damage. For stability theory it matters because it sets a floor on the thermal source term and because, in fast transients, a small additional decay-heat contribution adds to the energy that feedback must absorb.

4.2 The Way–Wigner correlation

A classic engineering estimate of decay power after an infinitely long irradiation is the Way–Wigner correlation,

$$\frac{P_d(t)}{P_0} \approx 0.066 \left[t^{-0.2} - (t + t_0)^{-0.2} \right], \tag{4.1}$$

where t is the time since shutdown and t_0 the operating time, both in seconds. For $t_0 \to \infty$ this reduces to $P_d/P_0 \approx 0.066\, t^{-0.2}$. Figure 4.1 plots this approximation. The slow, power-law decline—a consequence of the broad distribution of fission-product half-lives—is the defining feature: there is no single time constant, and the heat lingers for a long time.

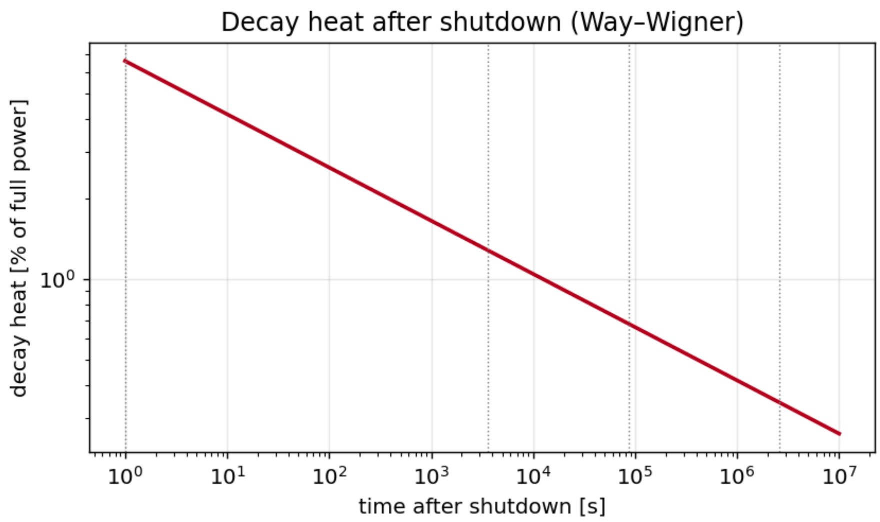

Figure 4.1: Decay heat as a fraction of full power following shutdown, from the Way–Wigner approximation (4.1). The power-law tail reflects the broad spectrum of fission-product half-lives; the heat cannot be turned off, only waited out and removed.

More accurate standards (ANSI/ANS-5.1) sum contributions from the major fissioning nuclides (^{235}U, ^{239}Pu, ^{238}U) with tabulated coefficients, and add the actinide contributions from ^{239}U and ^{239}Np, which are significant in the first day. For our purposes the Way–Wigner picture captures the essential physics: a large, slowly decaying, irreducible source.

4.3 Decay heat in the kinetics model

Decay heat can be appended to the point kinetics framework with its own set of decay-heat groups ω_k, in direct analogy with the precursors:

$$P_{\text{total}}(t) = P_f(t) + \sum_k \omega_k(t), \qquad \frac{\mathrm{d}\omega_k}{\mathrm{d}t} = \kappa_k P_f(t) - \lambda_k^{(d)} \omega_k(t), \tag{4.2}$$

where κ_k are the energy fractions and $\lambda_k^{(d)}$ the decay-heat group constants. In a fast power transient the fission term P_f dominates and decay heat is a small correction; in the aftermath of a scram it is the only term left. The companion software of Chapter 19 includes decay-heat groups for exactly this reason.

4.4 The thermal consequence: a worked estimate

Example 4.1 (Adiabatic heat-up after loss of cooling)**.** Suppose cooling is lost just after shutdown from full power P_0, and decay heat is initially $q = 0.06\,P_0$. If the core thermal mass is C and no heat is removed, the fuel temperature rises at

$$\frac{\mathrm{d}T_f}{\mathrm{d}t} = \frac{q}{C}. \tag{4.3}$$

For a core with $C \sim 10^7$ J/K and $P_0 = 3 \times 10^9$ W, the initial decay power is $q \approx 1.8 \times 10^8$ W, so $\mathrm{d}T_f/\mathrm{d}t \approx 18$ K/s. Starting near the normal operating temperature, the fuel would reach the melting point of UO_2 ($\sim 2800°C$) in a few minutes. This is the quantitative basis for the absolute requirement of emergency core cooling, and it explains why the *thermal* accident can proceed even after the *nuclear* reaction is stopped.

Key point. Stopping fission is necessary but not sufficient for safety. Decay heat is an irreducible thermal source of several per cent of full power that must be removed for days. Thermal stability concerns the fission transient; thermal *safety* also demands that the decay-heat path to the ultimate heat sink is never lost. Chernobyl began as a fission-power instability; its enduring danger came from the decay heat in the wrecked core.

4.5 Summary of Part I

We now have the foundations. Reactivity (2.3) measures distance from criticality; the six-factor formula (2.4) localises where feedback acts; the point kinetics equations (3.2) give the dynamics, with delayed neutrons setting the controllable timescale and the inhour equation (3.5) the period; the prompt-jump (3.4) captures the instantaneous response; and decay heat provides an irreducible thermal source. Part II turns to the thermal side: how the fission power becomes temperature, and how temperature, in turn, becomes reactivity—closing the feedback loop whose sign decides stability.

4.6 Decay heat from the fission-product inventory

The Way–Wigner law follows from a convolution of the operating history with the per-fission decay-power kernel. Let $f(\tau)$ be the decay power released at age τ after a single fission (W per fission). By linearity of radioactive decay, for a fission rate $R(t')$,

$$P_d(t) = \int_{-\infty}^{t} R(t')\, f(t-t')\, \mathrm{d}t' = \int_0^{\infty} R(t-\tau)\, f(\tau)\, \mathrm{d}\tau. \tag{4.4}$$

Proposition 4.2 (Way–Wigner from a power-law kernel)**.** *If $f(\tau) = C\,\tau^{-(1+a)}$ with $0 < a < 1$, and the reactor runs at constant fission rate R_0 for time T_0 before shutdown at $t = 0$, then for $t > 0$*

$$P_d(t) = \frac{CR_0}{a}\Big[\, t^{-a} - (t+T_0)^{-a} \Big]. \tag{4.5}$$

With $a = 0.2$ and normalisation $CR_0/(aE_f) \to 0.066\, P_0$ this is the Way–Wigner correlation; the limit $T_0 \to \infty$ gives $P_d/P_0 = 0.066\, t^{-0.2}$.

Proof. For $R(t') = R_0$ on $[-T_0, 0]$ and zero otherwise, the surviving contributions at $t > 0$ have ages $\tau \in [t, t+T_0]$, so $P_d(t) = R_0 \int_t^{t+T_0} C\tau^{-(1+a)} \mathrm{d}\tau = \frac{CR_0}{a}[t^{-a} - (t+T_0)^{-a}]$ via $\int \tau^{-(1+a)} \mathrm{d}\tau = -\tau^{-a}/a$. $\square$

The power-law kernel itself arises from the broad spread of fission-product decay constants: a density of modes $g(\lambda) \propto \lambda^{-1}$ per unit $\ln \lambda$ gives $f(\tau) = \int g(\lambda) E(\lambda) \lambda e^{-\lambda\tau} \mathrm{d}\lambda \propto \tau^{-1}$, steepened to $\tau^{-1.2}$ by the energy weighting.

4.7 The decay-heat group equations and their solution

Discretising the kernel into exponential groups is equivalent to the linear system

$$\frac{\mathrm{d}\omega_k}{\mathrm{d}t} = \kappa_k P_f(t) - \lambda_k^{(d)}\omega_k, \qquad P_d(t) = \sum_{k=1}^{K} \lambda_k^{(d)}\omega_k(t). \tag{4.6}$$

For constant power P_0 on $[-T_0, 0]$ the pre-shutdown inventory is $\omega_k(0) = \dfrac{\kappa_k P_0}{\lambda_k^{(d)}}\bigl(1 - e^{-\lambda_k^{(d)}T_0}\bigr)$, and after shutdown each group decays independently, giving

$$P_d(t) = \sum_k \kappa_k P_0\bigl(1 - e^{-\lambda_k^{(d)}T_0}\bigr)e^{-\lambda_k^{(d)}t}, \tag{4.7}$$

the standard ANSI/ANS-5.1 representation, whose continuum limit reproduces Proposition 4.2.

4.8 Adiabatic heat-up with time-dependent decay heat

Integrating the true (decreasing) source rather than a constant q, with $P_d(t) = 0.066\,P_0\,t^{-0.2}$ and $C\,\mathrm{d}T/\mathrm{d}t = P_d$,

$$\Delta T(t) = \frac{1}{C}\int_0^t P_d\,\mathrm{d}t' = \frac{0.066\,P_0}{0.8\,C}\,t^{0.8} = \frac{0.0825\,P_0}{C}\,t^{0.8}. \tag{4.8}$$

Corollary 4.3 (Time to a temperature limit)**.** *The adiabatic time to reach a rise* $\Delta T_{\max}$ *is* $t_{\max} = \bigl(0.8\,C\,\Delta T_{\max}/(0.066\,P_0)\bigr)^{1/0.8}$. *For* $P_0 = 3\times10^9\,\mathrm{W}$*,* $C = 10^7\,\mathrm{J/K}$*,* $\Delta T_{\max} = 2000\,\mathrm{K}$*, this gives* $t_{\max} \approx 1.3\times10^2\,\mathrm{s}$.

Exercises

1. Using the Way–Wigner approximation, estimate the decay-heat fraction at 1 s, 1 h, and 1 day after shutdown.
2. For a 3000 $\mathrm{MW_{th}}$ core with thermal mass $C = 10^7$ J/K and 6% initial decay heat, estimate the adiabatic heat-up rate just after a loss of cooling.
3. Why does decay heat decline as a power law rather than a single exponential?
4. Explain why stopping the fission chain reaction does not remove the need for cooling.
5. How does decay heat connect a reactivity transient to a thermal accident?

Part II

Heat Transport and Reactivity Feedback

CHAPTER 5

Heat Generation and Transport in the Core

Reactivity feedback acts through temperature, so we must know how fission power becomes temperature. This chapter follows the heat from its birth in the fuel, across the gap and cladding, to the coolant, and reduces the chain to the lumped thermal models used in stability analysis.

5.1 The path of the heat

Almost all fission energy—the kinetic energy of the fragments—is deposited in the fuel pellet within micrometres of the fission event. From there it must travel: by *conduction* through the ceramic fuel, across the (often gas-filled) *gap* between pellet and cladding, through the metal *cladding*, and finally by *convection* into the flowing coolant, which carries it out of the core to the steam generators or directly to the turbine. Each step imposes a temperature drop, and the sum of these drops sets the fuel centreline temperature for a given power.

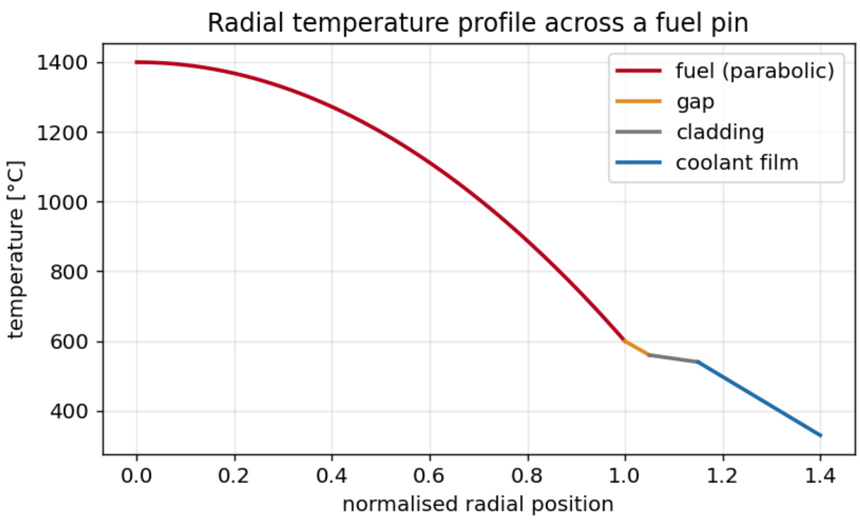

Figure 5.1: Schematic radial temperature profile across a fuel pin. The largest drop is the parabolic profile within the low-conductivity ceramic fuel; further drops occur across the gap, cladding, and coolant film. The fuel centreline runs hottest and responds essentially instantly to power changes—the basis of prompt Doppler feedback.

5.2 Conduction in the fuel pellet

For a cylindrical fuel pin with uniform volumetric heat generation q''' and constant conductivity k_f, the steady radial conduction equation

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\,k_f\frac{\mathrm{d}T}{\mathrm{d}r}\right)+q'''=0 \tag{5.1}$$

integrates to the parabolic profile

$$T(r)=T_s+\frac{q'''}{4k_f}\left(R^2-r^2\right), \tag{5.2}$$

so the centreline-to-surface temperature rise is

$$\Delta T_{\mathrm{fuel}}=T_c-T_s=\frac{q'''R^2}{4k_f}=\frac{q'}{4\pi k_f}, \tag{5.3}$$

where $q'=q'''\pi R^2$ is the linear heat rate (W/m). Remarkably, this rise depends only on the linear heat rate and the conductivity, not on the pin radius. Because UO_2 has a low and temperature-dependent conductivity (k_f falls from $\sim 8\,\mathrm{W/(m\,K)}$ near room temperature to $\sim 2\,\mathrm{W/(m\,K)}$ at operating temperature), the centreline runs very hot, often above 1000°C at rated power, even when the surface is near the coolant temperature. The integral conductivity $\int k_f\,\mathrm{d}T$ is the natural variable and is tabulated for fuel design.

5.3 Gap, cladding, and film

Across the pellet–clad gap, heat crosses by conduction through the fill gas and by radiation; the effective gap conductance h_g depends sensitively on the gap width, gas composition (helium fill, fission-gas dilution), and burnup. The drop is $\Delta T_g=q'/(2\pi R\,h_g)$. Conduction through the thin metal cladding adds a small drop. Finally, convection into the coolant follows Newton's law of cooling with a heat-transfer coefficient h,

$$\Delta T_{\mathrm{film}}=\frac{q'}{2\pi R_{co}\,h}, \tag{5.4}$$

where h is obtained from correlations (Dittus–Boelter for single-phase forced convection; more elaborate correlations in boiling). The total drop from centreline to bulk coolant is the sum of all four contributions.

5.4 The decisive timescales

For dynamics, the *rates* of temperature change matter as much as the steady drops. The fuel has a thermal time constant

$$\tau_f=\frac{m_f c_{p,f}}{h_{\mathrm{eff}}A}, \tag{5.5}$$

typically a few seconds, after which it equilibrates with the coolant for a given power. But the *centreline* responds to a power change essentially instantly—within the conduction time across the pellet—which is why Doppler feedback (which depends on fuel temperature) is treated as prompt. The coolant has its own transport time, the time for fluid to traverse the core, of order a second; moderator and coolant feedback therefore lag the power by this transport delay. These differing lags—prompt fuel, delayed coolant—are central to stability: a fast *negative* fuel feedback can stabilise a transient that a delayed coolant feedback could not catch in time.

> **Key point.** The fuel temperature responds promptly to power; the coolant temperature and void respond with a transport delay. Prompt negative feedback (Doppler) is the reactor's first and fastest line of defence in a fast transient; delayed feedback governs the slower stability of the operating point.

5.5 Lumped thermal models for dynamics

Full thermal-hydraulics is a field in itself, but stability analysis usually uses *lumped* models that capture the essential dynamics with a few ordinary differential equations. A widely used two-node model treats the fuel and coolant as single nodes with temperatures T_f and T_c:

$$m_f c_f \frac{\mathrm{d}T_f}{\mathrm{d}t} = P(t) - UA\,(T_f - T_c), \tag{5.6}$$

$$m_c c_c \frac{\mathrm{d}T_c}{\mathrm{d}t} = UA\,(T_f - T_c) - 2\dot{m}c_c(T_c - T_{in}), \tag{5.7}$$

where P is the power deposited in the fuel, UA the fuel-to-coolant conductance, $\dot{m}$ the coolant mass flow, and T_{in} the inlet temperature. The factor of two in the last term arises from taking the node temperature as the average of inlet and outlet. Even simpler is the *adiabatic* model used in fast-excursion analysis, in which heat removal is neglected on the (short) excursion timescale and the fuel simply integrates the power:

$$C\frac{\mathrm{d}T_f}{\mathrm{d}t} = P(t), \tag{5.8}$$

with C the fuel heat capacity. This is the model behind the Nordheim–Fuchs analysis of Chapter 9 and the feedback demonstrations of Chapter 19. Its virtue is that, in the seconds of a prompt excursion, it is also physically accurate: there simply is not time to remove the heat.

These lumped temperatures are the inputs to the feedback coefficients of the next chapter. Their dynamics—how fast T_f and T_c track power—combine with the *signs* of the feedback coefficients to determine whether the closed loop is stable.

5.6 Exact conduction in the fuel pin

Integral conductivity. With temperature-dependent k_f, define $\theta(T) = \int_{T_{\mathrm{ref}}}^{T} k_f(T')\mathrm{d}T'$. Integrating the steady equation $\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}(rk_f\,T') + q''' = 0$ once gives $k_f\,\mathrm{d}T = -(q'''/2)\,r\,\mathrm{d}r$, hence

$$\theta(T_c) - \theta(T_s) = \frac{q'''R^2}{4} = \frac{q'}{4\pi}, \tag{5.9}$$

independently of the form of $k_f(T)$: it is the integral conductivity, not the temperature, that is linear in the linear heat rate q'.

Exact transient solution. The homogeneous cylindrical heat equation $\partial_t T = \kappa(\partial_{rr}T + r^{-1}\partial_r T)$, $\kappa = k_f/(\rho c_p)$, with $T(R,t) = T_s$, $\partial_r T(0,t) = 0$, separates as $T - T_s = \mathcal{R}(r)e^{-\kappa\nu^2 t}$ with $\mathcal{R}'' + \mathcal{R}'/r + \nu^2\mathcal{R} = 0$, regular solution $J_0(\nu r)$. The condition $J_0(\nu R) = 0$ quantises $\nu_n = \alpha_n/R$ ($J_0(\alpha_n) = 0$; $\alpha_1 \approx 2.405$), giving

$$T(r,t) - T_s = \sum_{n=1}^{\infty} A_n J_0\Big(\frac{\alpha_n r}{R}\Big) \exp\Big(-\frac{\kappa\alpha_n^2}{R^2}t\Big), \tag{5.10}$$

with A_n from the orthogonality $\int_0^R J_0(\frac{\alpha_n r}{R})J_0(\frac{\alpha_m r}{R})r\,\mathrm{d}r = \frac{1}{2}R^2 J_1(\alpha_n)^2\delta_{nm}$.

Proposition 5.1 (Fundamental thermal time constant)**.** *For $t \gtrsim R^2/\kappa$ the series* (5.10) *is dominated by $n = 1$, so the fuel relaxes as a single exponential with*

$$\tau_f = \frac{R^2}{\kappa\,\alpha_1^2} = \frac{\rho c_p R^2}{k_f\,\alpha_1^2} \approx \frac{\rho c_p R^2}{5.78\,k_f}. \tag{5.11}$$

Proof. Mode n decays at $\kappa\alpha_n^2/R^2$, increasing in n; the ratio of mode 2 to mode 1 is $e^{-\kappa(\alpha_2^2-\alpha_1^2)t/R^2} \to 0$, leaving the $n = 1$ rate $\kappa\alpha_1^2/R^2$, i.e. $\tau_f = R^2/(\kappa\alpha_1^2)$, $\alpha_1^2 \approx 5.78$. □

The Biot number. With $\mathrm{Bi} = hR/k_f$ comparing conduction to convection resistance, the lumped balance $\rho c_p V\,\mathrm{d}T/\mathrm{d}t = -hA(T - T_\infty)$ (decay constant $\tau = \rho c_p V/(hA)$) is valid for $\mathrm{Bi} \ll 1$; for $\mathrm{Bi} \gtrsim 1$, typical of ceramic fuel, the distributed solution (5.10) and Proposition 5.1 are required.

5.7 Spectral analysis of the two-node thermal model

Linearising the two-node model about steady state gives $\dot{\boldsymbol{\delta}} = \mathbf{B}\boldsymbol{\delta}$ for $(\delta T_f, \delta T_c)$ with

$$\mathbf{B} = \begin{pmatrix} -\dfrac{UA}{m_f c_f} & \dfrac{UA}{m_f c_f} \\ \dfrac{UA}{m_c c_c} & -\dfrac{UA + 2\dot{m}c_c}{m_c c_c} \end{pmatrix}. \tag{5.12}$$

Here $\operatorname{tr}\mathbf{B} < 0$ and $\det\mathbf{B} = \dfrac{UA\cdot 2\dot{m}c_c}{m_f c_f\, m_c c_c} > 0$, so by the $n = 2$ Routh–Hurwitz conditions (Theorem 8.2) both eigenvalues

$$\lambda_\pm = \tfrac{1}{2}\Big(\operatorname{tr}\mathbf{B} \pm \sqrt{\operatorname{tr}^2\mathbf{B} - 4\det\mathbf{B}}\Big) \tag{5.13}$$

lie in the left half-plane: the passive thermal network is unconditionally stable. The constants $\tau_\pm = -1/\lambda_\pm$ are the fast (fuel) and slow (coolant) lags entering the feedback transfer function $H(s)$ of Chapter 8. Instability can thus enter only through the neutronic coupling and the sign of the reactivity feedback, never through the thermal network itself.

Exercises

1. Show that the fuel centreline-to-surface temperature rise depends only on the linear heat rate and conductivity, not the pin radius.

2. List the four temperature drops from fuel centreline to bulk coolant and identify which is usually largest.
3. Explain why Doppler feedback is treated as prompt while moderator feedback is delayed.
4. Justify the adiabatic fuel model (5.8) for analysing a prompt excursion.
5. Estimate the fuel thermal time constant given plausible values and discuss its role in stability.

CHAPTER 6

Reactivity Feedback Mechanisms

Feedback is where thermal physics re-enters the neutron economy. This chapter defines the feedback coefficients, derives the dominant mechanisms—Doppler, moderator/coolant temperature, and void—and establishes the signs that decide stability. The void coefficient receives special attention because its sign is the crux of the Chernobyl story.

6.1 Temperature coefficients of reactivity

A **temperature coefficient of reactivity** is the change in reactivity per unit change in some temperature,

$$\alpha_T = \frac{\partial \rho}{\partial T}. \tag{6.1}$$

There is one for each material whose temperature affects the neutron economy: $\alpha_D = \partial\rho/\partial T_f$ (the fuel or Doppler coefficient), $\alpha_M = \partial\rho/\partial T_M$ (the moderator coefficient), and analogously for coolant and structure. For stability, the sign is paramount: a *negative* coefficient means that heating removes reactivity, which opposes a power rise—stabilising. A *positive* coefficient means heating adds reactivity, reinforcing a power rise—destabilising.

Definition 6.1 (Power coefficient)**.** The **power coefficient** $\alpha_P = \partial\rho/\partial P$ aggregates all temperature feedbacks weighted by how strongly each temperature responds to power:

$$\alpha_P = \alpha_D \frac{\partial T_f}{\partial P} + \alpha_M \frac{\partial T_M}{\partial P} + \alpha_v \frac{\partial \bar{v}}{\partial P} + \cdots \tag{6.2}$$

A negative power coefficient is the single most important guarantee of inherent stability: it means that any increase in power is self-limiting. Western power reactors are designed and licensed to have a negative power coefficient across the operating range.

6.2 The Doppler effect: prompt and reliably negative

The Doppler coefficient is the reactor's fastest and most dependable feedback. It arises from the broadening of the neutron-capture resonances of ^{238}U (and ^{240}Pu) as the fuel heats.

At a resonance, the absorption cross-section is sharply peaked at a resonance energy E_0. The thermal motion of the absorbing nuclei—greater at higher temperature—Doppler-shifts the relative neutron energy, smearing the resonance into a broader, lower peak while approximately conserving its area (Figure 6.1). The key consequence is that broadening *reduces self-shielding*: when the resonance is narrow and tall, neutrons at the resonance energy are absorbed in the thin outer layer of the fuel and the interior is "shielded"; when the resonance is broadened, more of the fuel volume participates in absorption. Net resonance absorption therefore *increases* with temperature, the resonance escape probability p *falls*, and reactivity *decreases*.

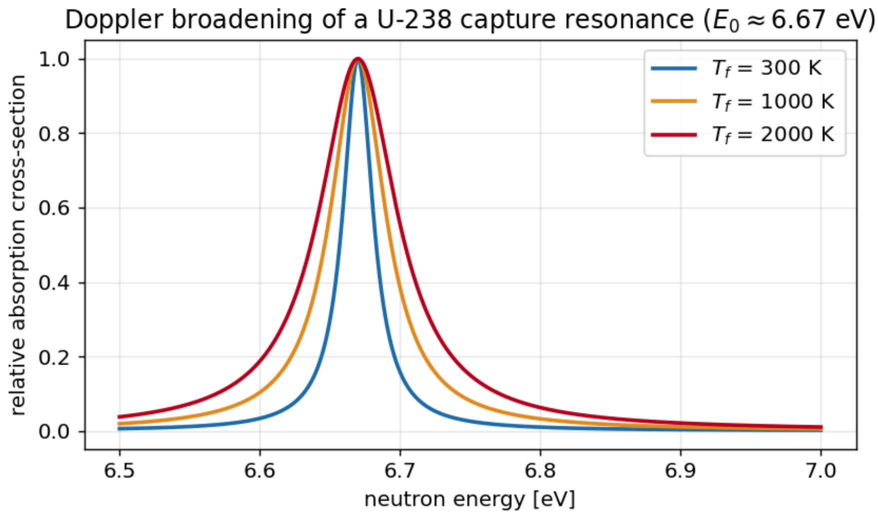

Figure 6.1: Doppler broadening of a ^{238}U capture resonance with increasing fuel temperature. The peak lowers and widens at roughly constant area; reduced self-shielding increases net resonance absorption, lowering reactivity. This is the prompt, negative Doppler feedback.

Two features make Doppler feedback uniquely valuable. First, it is *prompt*: it depends on the fuel temperature, which rises essentially instantly with power, so it acts even on a prompt-critical excursion. Second, it is *reliably negative* in any fuel containing a fertile resonance absorber such as ^{238}U, which is to say in essentially all power-reactor fuel. A useful approximate form is

$$\alpha_D \approx \frac{A}{\sqrt{T_f}}, \qquad \Delta\rho_D = \int \alpha_D \, \mathrm{d}T_f \propto \sqrt{T_f}, \tag{6.3}$$

with $A < 0$; the $1/\sqrt{T_f}$ dependence means the coefficient weakens at high temperature but never changes sign. It was the absence of a sufficiently strong *net* prompt negative feedback, not the absence of Doppler per se, that doomed the RBMK: the RBMK had Doppler feedback, but at low power its huge positive void coefficient overwhelmed it.

Key point. Doppler feedback is prompt and negative. It is the only feedback fast enough to limit a prompt-critical excursion, and it is the physical reason a well-designed reactor “rides out” a reactivity insertion. The decisive question for any reactor is whether the prompt feedback is *net* negative once the void coefficient is included.

6.3 Moderator and coolant temperature feedback

In a thermal reactor the moderator slows neutrons to thermal energies. Its temperature affects reactivity through two competing routes: changes in moderator *density* (a hotter, less dense moderator slows neutrons less effectively), and changes in the thermal neutron *spectrum* (a hotter moderator hardens the spectrum, shifting absorption rates). In a light-water reactor the density effect dominates, and because water is both moderator and coolant, heating it reduces moderation. Whether this raises or lowers reactivity depends on whether the lattice is under- or over-moderated:

- In an **under-moderated** lattice (PWR design intent), there is "too little" moderation at the optimum; reducing it further (by heating/expanding the water) lowers reactivity. The moderator temperature coefficient (MTC) is *negative*—stabilising.
- In an **over-moderated** lattice, reducing moderation moves toward the optimum and *raises* reactivity. The MTC is positive—destabilising.

Light-water reactors are deliberately designed under-moderated to guarantee a negative MTC. The moderator coefficient acts with the coolant transport delay and so governs the slower stability of the operating point rather than the prompt excursion.

6.4 The void coefficient

When coolant boils (in a BWR, or in any reactor during a transient) or flashes to steam, it forms *voids*—regions of vapour with much lower density than liquid. The **void coefficient** $\alpha_v = \partial\rho/\partial\bar{v}$, where $\bar{v}$ is the void fraction, captures the reactivity effect. Voiding has several effects at once: it removes moderator (in a water-moderated reactor), it removes a neutron absorber (water absorbs neutrons), and it changes leakage. The net sign depends, once again, on whether the lattice is under- or over-moderated.

- In an **under-moderated** water reactor (PWR, BWR), the loss of moderation dominates: voiding lowers reactivity, $\alpha_v < 0$—stabilising. A BWR runs with boiling by design, and its negative void coefficient is the basis of its self-regulation (and, in fast transients, of a mild instability tendency discussed in Chapter 8).
- In the **RBMK**, moderation was provided primarily by *graphite*, which is unaffected by boiling the water. The water acted mainly as coolant and as a parasitic *absorber*. Voiding the water therefore removed an absorber without removing much moderation, so reactivity *rose*: $\alpha_v > 0$, strongly so, and most strongly at low power where the steam fraction was small and sensitive. This is the single most important design fact behind Chernobyl.

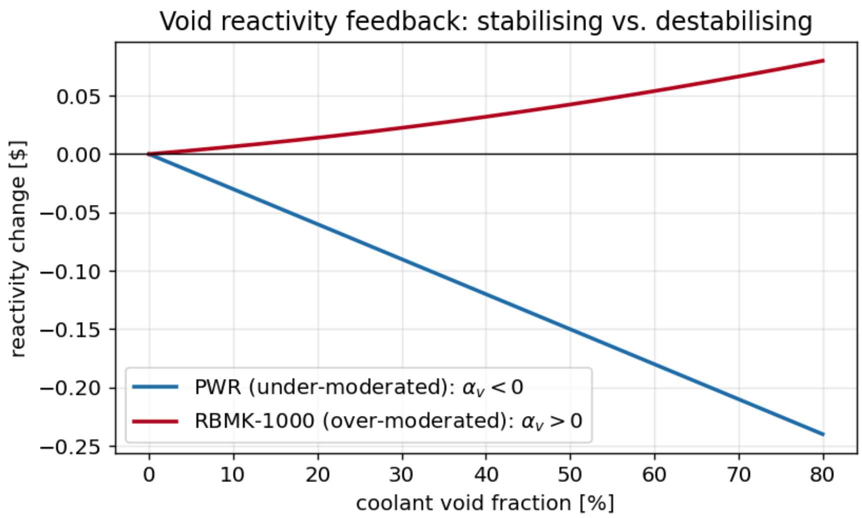

Figure 6.2: Void reactivity feedback for an under-moderated lattice (PWR, negative slope) and an over-moderated lattice (RBMK-1000, positive slope). The void coefficient is the slope at the operating point; its sign decides whether boiling is self-limiting or self-amplifying.

Remark 6.2 (The vicious circle of a positive void coefficient). A positive void coefficient creates a runaway loop: power up → more boiling → more voids → *more* reactivity → more power. If the prompt Doppler feedback is not strong enough to overcome it, the loop is unstable. At the RBMK's low-power, low-flow state on the night of the accident, the loop gain exceeded unity and the prompt feedback was net positive—a condition the machine should never have been allowed to enter.

6.5 Other feedbacks: expansion, bowing, and burnup

Several smaller effects round out the picture. *Thermal expansion* of fuel, cladding, and structure changes geometry and hence leakage and lattice spacing; in fast reactors, fuel and structural expansion provide important negative feedback. *Fuel bowing* and *control-rod-bank expansion* can contribute small coefficients. Over the long term, *burnup* changes the isotopic composition and hence all coefficients—a reactor's MTC, for example, becomes more negative as soluble boron is reduced over a cycle. For the fast dynamics of stability, however, the trio of Doppler, moderator, and void dominate, and the contest between prompt-negative Doppler and a possibly-positive void coefficient is the essential drama.

6.6 Assembling the feedback reactivity

For the coupled model we write the total reactivity as the externally imposed part plus the feedback,

$$\rho(t) = \rho_{ext}(t) + \alpha_D\,[T_f(t) - T_f^0] + \alpha_M\,[T_c(t) - T_c^0] + \alpha_v\,[\bar{v}(t) - \bar{v}^0] + \cdots \tag{6.4}$$

Substituting (6.4) into the point kinetics (3.2) and coupling to the thermal equations of Chapter 5 closes the loop. The resulting nonlinear system is the subject of Chapter 7 and of the stability analysis of Part III. Its behaviour hinges entirely on the signs and magnitudes of the coefficients in (6.4): a system with net-negative, suitably prompt feedback is stable and self-regulating; a system with net-positive feedback, especially a prompt one, is an accident waiting for a trigger.

6.7 Rigorous theory of the Doppler coefficient

We derive the temperature dependence of the Doppler coefficient from resonance-absorption theory. Near an isolated resonance at E_0 the capture cross-section of a *stationary* target is the Breit–Wigner form

$$\sigma_\gamma^0(E) = \sigma_0\,\frac{\Gamma_\gamma}{\Gamma}\sqrt{\frac{E_0}{E}}\,\frac{1}{1+x^2}, \qquad x = \frac{2(E-E_0)}{\Gamma}, \tag{6.5}$$

with total width Γ, radiative width Γ_γ, and peak σ_0. Thermal motion of the target nuclei convolves this with a Maxwellian of relative velocities, giving the Doppler-broadened shape

$$\sigma_\gamma(E,T) = \sigma_0\,\frac{\Gamma_\gamma}{\Gamma}\sqrt{\frac{E_0}{E}}\,\psi(\xi,x), \qquad \psi(\xi,x) = \frac{\xi}{2\sqrt{\pi}}\int_{-\infty}^{\infty}\frac{e^{-\frac{1}{4}\xi^2(x-y)^2}}{1+y^2}\,\mathrm{d}y, \tag{6.6}$$

with broadening parameter $\xi = \Gamma/\Delta_D$ and *Doppler width*

$$\Delta_D = \sqrt{\frac{4E_0 k_B T}{A}} \;\propto\; \sqrt{T}, \tag{6.7}$$

A the target mass number. The convolution (6.6) conserves area, $\int_{-\infty}^{\infty} \psi \, \mathrm{d}x = \pi$, so broadening lowers and widens the line without changing its integral.

The reactivity effect operates through the *effective resonance integral* under self-shielding. In the narrow-resonance approximation with background cross-section σ_b per absorber atom,

$$I_{\text{eff}}(T) = \int \frac{\sigma_\gamma(E,T)\,\sigma_b}{\sigma_\gamma(E,T) + \sigma_b} \frac{\mathrm{d}E}{E}, \tag{6.8}$$

which is an increasing function of the Doppler width.

Proposition 6.3 (Temperature law of the Doppler coefficient)**.** *To leading order in the broadening, $\partial I_{\text{eff}}/\partial T > 0$ with $\partial I_{\text{eff}}/\partial T \propto \mathrm{d}\Delta_D/\mathrm{d}T \propto T^{-1/2}$. Consequently the Doppler coefficient obeys*

$$\alpha_D(T) = \frac{\partial \rho}{\partial T} = -\frac{1}{p}\frac{\partial p}{\partial I_{\text{eff}}}\frac{\partial I_{\text{eff}}}{\partial T} = \frac{A_D}{\sqrt{T}}, \qquad A_D < 0, \tag{6.9}$$

so α_D is negative for all T and weakens as $T^{-1/2}$ without ever changing sign. The reactivity defect $\Delta\rho_D(T) = \int_{T_0}^{T} \alpha_D \, \mathrm{d}T' \propto -(\sqrt{T} - \sqrt{T_0})$ is strictly negative and decreasing.

Proof. Self-shielding makes I_{eff} increase as the line broadens because the saturated (flat-topped) region of the integrand widens; differentiating the narrow-resonance integral and using $\partial\Delta_D/\partial T = \frac{1}{2}\Delta_D/T \propto T^{-1/2}$ from (6.7) gives $\partial I_{\text{eff}}/\partial T \propto T^{-1/2}$. Increased resonance capture lowers the resonance escape probability, $\partial p/\partial I_{\text{eff}} < 0$, so $\alpha_D = -p^{-1}(\partial p/\partial I_{\text{eff}})(\partial I_{\text{eff}}/\partial T)$ is a negative constant times $T^{-1/2}$. Integrating yields the stated defect. □

Proposition 6.3 is the rigorous content of the earlier heuristic: Doppler feedback is prompt (it depends only on T_f), structurally negative (it needs only a fertile resonance absorber such as ^{238}U), and bounded below by a $T^{-1/2}$ law.

Exercises

1. Define the power coefficient and explain why a negative power coefficient guarantees self-limitation.
2. Explain the mechanism by which Doppler broadening reduces self-shielding and lowers reactivity.
3. For an under- and an over-moderated lattice, predict the sign of the void coefficient and justify it.
4. Why is the void coefficient of the RBMK largest at low power?
5. Write the total feedback reactivity (6.4) and identify which term is prompt.

CHAPTER 7

Coupled Neutronic–Thermal-Hydraulic Dynamics

We now assemble the pieces—point kinetics, lumped thermal model, and feedback coefficients—into a single coupled dynamical system and explore its qualitative behaviour before the formal stability analysis of Part III.

7.1 The closed-loop model

Combining the point kinetics (3.2), the lumped thermal model of Chapter 5, and the feedback law (6.4), a compact representative model is

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{\rho(t) - \beta}{\Lambda}\,p + \sum_i \lambda_i \zeta_i, \tag{7.1}$$

$$\frac{\mathrm{d}\zeta_i}{\mathrm{d}t} = \frac{\beta_i}{\Lambda}\,p - \lambda_i \zeta_i, \tag{7.2}$$

$$C_f \frac{\mathrm{d}T_f}{\mathrm{d}t} = P_0\,p - UA\,(T_f - T_c), \tag{7.3}$$

$$C_c \frac{\mathrm{d}T_c}{\mathrm{d}t} = UA\,(T_f - T_c) - 2\dot{m}c_c(T_c - T_{in}), \tag{7.4}$$

$$\rho(t) = \rho_{ext}(t) + \alpha_D(T_f - T_f^0) + \alpha_c(T_c - T_c^0). \tag{7.5}$$

This is a system of nonlinear ordinary differential equations: nonlinear because reactivity multiplies power in the first equation, and reactivity itself depends on the temperatures driven by power. The nonlinearity is essential—it is what makes a power excursion turn over rather than grow forever—but it also makes the behaviour rich.

7.2 The self-regulating reactor in words

Before any mathematics, the qualitative behaviour of a *negative-feedback* reactor is worth stating plainly. Suppose the operator withdraws a control rod, inserting positive ρ_{ext}. Power rises. The fuel heats within milliseconds; if $\alpha_D < 0$, this immediately subtracts reactivity, blunting the rise. As the coolant heats over the next second, $\alpha_c < 0$ subtracts more. The reactor settles at a higher power and temperature at which the feedback exactly cancels the inserted reactivity: a new stable operating point. The reactor has *regulated itself* without any control action. Conversely, if the operator reduces power, temperatures fall, feedback adds reactivity, and the fall is arrested. This is the everyday meaning of a negative power coefficient.

Now reverse the sign of the dominant feedback, as in the RBMK at low power. The rod withdrawal raises power; boiling increases; the positive void coefficient adds *more* reactivity; power rises further; and unless a strong prompt negative effect intervenes, the excursion accelerates. The same control action that a stable reactor absorbs, the unstable one amplifies.

7.3 Equilibrium and the power defect

At equilibrium the time derivatives vanish and the feedback exactly offsets the external reactivity. The total reactivity that must be supplied by control to hold the reactor critical from cold-zero-power to hot-full-power is the **power defect**,

$$\Delta\rho_{\text{defect}} = -\int_0^{P_0} \alpha_P \, \mathrm{d}P, \tag{7.6}$$

which for a negative power coefficient is positive (control must *add* reactivity to reach full power, because the feedback removes it). The power defect is a useful integral measure of how strongly self-regulating a reactor is, and it must be matched by available control-rod and chemical-shim worth.

7.4 Linearisation preview

For small perturbations about an operating point, write $p = p_0 + \delta p$, $T_f = T_f^0 + \delta T_f$, etc., and keep first-order terms. The linearised system has the form $\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{b}\,\rho_{ext}$, and its stability is decided by the eigenvalues of the matrix $\mathbf{A}$: the operating point is stable if and only if all eigenvalues have negative real part. The feedback coefficients enter $\mathbf{A}$ directly, so the sign and magnitude of α_D, α_c, and α_v determine the eigenvalues' signs. This is the formal content of Chapter 8, where we also pass to the frequency domain and the Nyquist criterion, which handles the transport delays more naturally than eigenvalues alone.

7.5 A demonstration of self-limitation

Figure 3.2 (bottom right) already showed the canonical result: a reactivity-initiated excursion that, with negative feedback, rises to a peak and then turns over as the heating fuel removes reactivity. The peak power, the time to peak, and the total energy released are set by the competition between the inserted reactivity and the feedback strength—precisely the Nordheim–Fuchs balance derived in Chapter 9. The same model with the feedback sign flipped does not turn over; it diverges until the model's assumptions break. The entire safety case of an inherently stable reactor rests on getting that sign right and the magnitude large enough.

> **Key point.** The coupled model is nonlinear, but its stability is governed by a simple principle: the feedback must remove reactivity as power and temperature rise, and it must do so promptly enough. Part III makes "promptly enough" precise through eigenvalues and the Nyquist criterion; Part IV shows what the violation of this principle looked like in practice.

Exercises

1. Describe in words the self-regulating response of a negative-feedback reactor to a rod withdrawal.
2. Define the power defect and explain its sign for a negative power coefficient.
3. Explain how the feedback coefficients enter the linearised system matrix and why feedback is more potent at higher power.
4. Sketch the qualitative power response of a reactor with positive feedback to a small rod withdrawal.
5. Why does a nonlinear feedback model produce a power excursion that turns over, while a linear analysis cannot capture the turnover?

Part III

Stability Theory

CHAPTER 8

Linear Stability and Frequency-Domain Methods

Stability is, in the first instance, a question about *small* perturbations: does a tiny disturbance grow or decay? This is the domain of linear stability analysis. We linearise the coupled model, introduce the zero-power and feedback transfer functions, and apply the Nyquist criterion, which handles the transport delays of real reactors more gracefully than eigenvalues alone.

8.1 Linearisation about an operating point

Let the reactor sit at steady power p_0 with steady temperatures, and perturb: $p = p_0 + \delta p$, $\zeta_i = \zeta_{i0} + \delta\zeta_i$, $T_f = T_f^0 + \delta T_f$, and so on. Substituting into the coupled equations of Chapter 7 and discarding products of small quantities gives a linear system

$$\frac{\mathrm{d}}{\mathrm{d}t}\,\delta\mathbf{x} = \mathbf{A}\,\delta\mathbf{x} + \mathbf{b}\,\delta\rho_{ext}, \tag{8.1}$$

where $\delta\mathbf{x} = (\delta p, \delta\zeta_1, \ldots, \delta\zeta_6, \delta T_f, \delta T_c)^\top$. The crucial linearised coupling is in the power equation: the term $(\rho-\beta)p/\Lambda$ contributes $p_0\,\delta\rho/\Lambda$, and $\delta\rho = \alpha_D\,\delta T_f + \alpha_c\,\delta T_c$. Thus the feedback coefficients enter $\mathbf{A}$ multiplied by p_0/Λ: *feedback is more potent at higher power*, an observation with direct bearing on why the RBMK was most dangerous at low power.

The operating point is asymptotically stable if and only if every eigenvalue of $\mathbf{A}$ has negative real part. For the bare point-kinetics block (no feedback) the eigenvalues are exactly the inhour roots of Chapter 3; adding negative feedback pushes the dominant root further into the left half-plane (more stable), while positive feedback pulls it toward, and possibly across, the imaginary axis (instability).

8.2 The zero-power transfer function

Frequency-domain analysis is more illuminating for systems with delays. Laplace transform the point kinetics about p_0 with input $\delta\rho$ and output $\delta p/p_0$; eliminating the precursors gives the **zero-power reactor transfer function**

$$Z(s) = \frac{\delta p/p_0}{\delta\rho} = \frac{1}{\Lambda s + \displaystyle\sum_i \frac{\beta_i s}{s+\lambda_i}}. \tag{8.2}$$

Its denominator is exactly the inhour function (3.5); its poles are the inhour roots. The name "zero-power" signals that no thermal feedback is included: it is the open-loop response of the neutronics alone. At low frequency $Z(s)$ behaves like an integrator with a long effective time constant set by the precursors; at high frequency $Z(s) \to 1/(\Lambda s)$, the fast prompt response. The transition between these regimes, around the delayed-neutron frequencies, is where the reactor's dynamic character lives.

8.3 Closing the loop: the feedback transfer function

Let the feedback path have transfer function $H(s)$ relating a power perturbation to the feedback reactivity it produces, $\delta\rho_{fb} = -H(s)\,\delta p/p_0$ (the minus sign for negative feedback). For a single lumped fuel node with coefficient $\alpha_D < 0$ and thermal time constant τ_f,

$$H(s) = \frac{|\alpha_D|\,K}{1 + \tau_f s}, \tag{8.3}$$

a first-order lag: fast feedback at low frequency, attenuated and phase-lagged at high frequency. The closed-loop transfer function from external reactivity to power is

$$G(s) = \frac{Z(s)}{1 + Z(s)H(s)}, \tag{8.4}$$

and stability is governed by the roots of the characteristic equation $1 + Z(s)H(s) = 0$. The product $Z(s)H(s)$ is the *open-loop gain*; everything hinges on its behaviour.

8.4 The Nyquist criterion

The Nyquist criterion reads stability directly off a plot of the open-loop gain $Z(j\omega)H(j\omega)$ as ω runs over all frequencies. The closed loop is stable provided the Nyquist locus does not encircle the critical point -1 in the complex plane. Physically, encirclement of -1 means there is a frequency at which the feedback, after its phase lag, returns *in phase* with the disturbance and with gain exceeding unity—positive feedback at that frequency, hence growth.

This is where *phase lag* becomes dangerous even for nominally negative feedback. If the feedback is negative at zero frequency but accumulates enough phase lag (from thermal transport delays, coolant transit, etc.) that at some frequency it is delayed by half a cycle, it acts as *positive* feedback at that frequency. If the loop gain there exceeds one, the reactor oscillates with growing amplitude. Boiling-water reactors, with their void feedback transported by the coolant, are the classic example: they can exhibit lightly damped or even growing power–flow oscillations under certain low-flow, high-power conditions, and operating limits are drawn specifically to keep the decay ratio safely below one.

Definition 8.1 (Decay ratio)**.** The decay ratio is the ratio of the amplitudes of two successive peaks of an oscillatory response to a perturbation. A decay ratio below 1 indicates a damped, stable response; equal to 1 marks the stability boundary; above 1 the oscillation grows. BWR stability is monitored and licensed in these terms, typically with a limit well below unity.

8.5 Gain and phase margins

The robustness of stability is quantified by the *gain margin* (how much the loop gain could increase before the locus reaches -1) and the *phase margin* (how much additional phase lag could be tolerated). A reactor designed for inherent stability carries comfortable margins so that variations over the fuel cycle, across power levels, and under off-normal conditions never erode stability. The RBMK at low power had, in effect, negative margins in the relevant configuration: its dominant feedback was positive, so no amount of phase bookkeeping could save it—the locus enclosed -1 outright.

8.6 Power dependence and the low-power trap

Because feedback enters the loop gain multiplied by p_0, a reactor's stability character changes with power. A *positive*-feedback reactor like the RBMK is most unstable at *low* power: there the steam void fraction is small and most sensitive, so the void coefficient is largest, while the prompt Doppler feedback (which scales with the fuel-temperature changes that low power barely produces) is weakest. The result is a window of operation—low power, high positive void worth, depleted control margin—in which the machine is acutely unstable. The Chernobyl test drove the reactor straight into that window.

Key point. Linear stability is decided by the eigenvalues of the linearised system, or equivalently by whether the Nyquist locus of the open-loop gain $Z(s)H(s)$ encircles -1. Negative feedback with small phase lag keeps the locus clear; positive feedback, or negative feedback with excessive transport lag, can enclose -1 and produce growing oscillations or divergence. Stability margins must hold across all power levels and cycle conditions—a requirement the RBMK violated at low power.

8.7 State-space form and the Routh–Hurwitz criterion

To make "stability" fully explicit, reduce to the one-precursor, one-temperature model that retains the essential feedback physics:

$$\dot{n} = \frac{\rho - \beta}{\Lambda} n + \lambda C, \quad \dot{C} = \frac{\beta}{\Lambda} n - \lambda C, \quad \dot{T} = \frac{n}{\mu} - \theta T, \quad \rho = \rho_0 + \alpha T, \tag{8.5}$$

with heat capacity $\mu > 0$, cooling rate $\theta > 0$, and feedback coefficient α. Linearising about the critical equilibrium (n_0, C_0, T_0) with $\rho_0 = 0$ gives the Jacobian

$$\mathbf{A} = \begin{pmatrix} -\dfrac{\beta}{\Lambda} & \lambda & \dfrac{\alpha n_0}{\Lambda} \\ \dfrac{\beta}{\Lambda} & -\lambda & 0 \\ \dfrac{1}{\mu} & 0 & -\theta \end{pmatrix}, \tag{8.6}$$

whose characteristic polynomial is $\det(s\mathbf{I} - \mathbf{A}) = s^3 + a_1 s^2 + a_2 s + a_3$ with

$$a_1 = \lambda + \theta + \frac{\beta}{\Lambda}, \quad a_2 = \theta\Big(\lambda + \frac{\beta}{\Lambda}\Big) - \frac{\alpha n_0}{\Lambda\mu}, \quad a_3 = -\frac{\alpha n_0 \lambda}{\Lambda\mu}. \tag{8.7}$$

Theorem 8.2 (Routh–Hurwitz, cubic)**.** *The polynomial $s^3 + a_1 s^2 + a_2 s + a_3$ (real coefficients) has all roots in the open left half-plane if and only if $a_1 > 0$, $a_3 > 0$, and $a_1 a_2 > a_3$.*

Proposition 8.3 (Feedback sign decides stability)**.** *With $\lambda, \theta, \beta, \Lambda, \mu, n_0 > 0$, the equilibrium is asymptotically stable if and only if $\alpha < 0$.*

Proof. By (8.7), $a_1 > 0$ always. Now $a_3 = -\alpha n_0 \lambda/(\Lambda\mu)$, so $a_3 > 0 \iff \alpha < 0$. If $\alpha < 0$ then also $a_2 = \theta(\lambda + \beta/\Lambda) + |\alpha| n_0/(\Lambda\mu) > 0$, and

$$a_1 a_2 - a_3 = \underbrace{(\lambda + \theta + \tfrac{\beta}{\Lambda})\,\theta(\lambda + \tfrac{\beta}{\Lambda})}_{>0} + \frac{|\alpha| n_0}{\Lambda\mu}\Big[(\lambda + \theta + \tfrac{\beta}{\Lambda}) - \lambda\Big] = (\cdots) + \frac{|\alpha| n_0}{\Lambda\mu}\Big(\theta + \tfrac{\beta}{\Lambda}\Big) > 0,$$

so all three Routh–Hurwitz conditions of Theorem 8.2 hold and the equilibrium is asymptotically stable. Conversely, if $\alpha > 0$ then $a_3 < 0$, violating the necessary condition $a_3 > 0$, so at least one root lies in the closed right half-plane and the equilibrium is unstable. ($\alpha = 0$ is the neutral case, $a_3 = 0$, with a root at the origin.) □

Proposition 8.3 is the precise version of the book's thesis: with all thermal and kinetic parameters positive, *the sign of the feedback coefficient alone* decides linear stability. The RBMK's low-power state had the effective $\alpha > 0$ and was therefore linearly unstable.

Lyapunov viewpoint. By Theorem E.2 (Appendix E), $\alpha < 0$ guarantees a quadratic Lyapunov function $V(\mathbf{x}) = \mathbf{x}^\top \mathbf{P}\mathbf{x}$ with $\mathbf{P} \succ 0$ solving $\mathbf{A}^\top \mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}$ for any $\mathbf{Q} \succ 0$; then $\dot{V} = -\mathbf{x}^\top \mathbf{Q}\mathbf{x} < 0$ certifies decay of every small perturbation. Stability is thus not merely spectral but energy-dissipative.

8.8 The Nyquist criterion, precisely

For the full delayed system the loop gain $L(s) = Z(s)H(s)$ is meromorphic, and stability is read off by the argument principle.

Theorem 8.4 (Nyquist). *Let $L(s)$ be the open-loop transfer function with P poles in the open right half-plane. As s traverses the Nyquist contour (the imaginary axis closed by an infinite right semicircle), let N be the number of clockwise encirclements of the point -1 by $L(s)$. Then the number of closed-loop poles in the right half-plane is $Z = N + P$. The closed loop is stable iff $Z = 0$, i.e. iff $N = -P$.*

Sketch. Closed-loop poles are zeros of $1 + L(s)$. By the argument principle, as s runs over the closed contour, the net change in $\arg(1 + L(s))$ equals $2\pi(Z - P)$, where Z, P count zeros and poles of $1 + L$ (equivalently poles of L) inside the contour. The change in $\arg(1 + L)$ is 2π times the clockwise encirclements of the origin by $1 + L$, i.e. of -1 by L; hence $N = Z - P$. □

For an open-loop-stable reactor ($P = 0$) the criterion reduces to the familiar statement: stability iff the locus $L(j\omega)$ does not encircle -1. Phase lag in $H(s)$ from thermal transport can rotate the locus until it encircles -1, which is the frequency-domain face of Proposition 8.3's sign change.

Exercises

1. Write the zero-power transfer function (8.2) and identify its poles.
2. State the Nyquist stability criterion and interpret encirclement of -1 physically.

3. Explain how phase lag can turn nominally negative feedback into destabilising feedback at some frequency.
4. Define the decay ratio and state the stability condition in terms of it.
5. Explain why a positive-feedback reactor is most unstable at low power.

CHAPTER 9

Nonlinear Power Excursions

Linear analysis tells us whether an operating point is stable. When a reactor is driven hard—a large, fast reactivity insertion—the response is fully nonlinear, and the question becomes: how high does the power spike, how much energy is deposited, and what stops it? The classic answer is the Nordheim–Fuchs model, which gives a remarkably clean closed-form description of a self-limiting prompt excursion. This is the model that quantifies what a *thermally stable* reactor does when subjected to a reactivity-initiated accident—and, by its failure, what Chernobyl did.

9.1 The reactivity-initiated accident

A reactivity-initiated accident (RIA) is a rapid insertion of positive reactivity—an ejected control rod, a cold-water slug, or, at Chernobyl, the catastrophic positive scram. If the insertion exceeds one dollar, the reactor becomes prompt-critical and the power rises on the Λ timescale, far too fast for delayed neutrons or mechanical systems to matter. The only thing that can arrest it is prompt feedback. Whether the excursion is a benign, self-terminating spike or an explosive energy release depends entirely on the sign and strength of that prompt feedback.

9.2 The Nordheim–Fuchs model

Consider a step insertion $\rho_0 > \beta$ on a prompt-critical reactor with prompt negative feedback. On the prompt timescale the delayed neutrons are a nearly constant background, so we drop the precursor term and write the prompt kinetics

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{\rho(t) - \beta}{\Lambda}\, p, \tag{9.1}$$

with feedback through an adiabatically heating fuel (Chapter 5),

$$\rho(t) = \rho_0 - \gamma\left(E(t)\right), \qquad \frac{\mathrm{d}E}{\mathrm{d}t} = p, \tag{9.2}$$

where $E(t) = \int_0^t p\,\mathrm{d}t'$ is the energy released and $\gamma > 0$ is the magnitude of the (negative) feedback per unit energy—the product of the fuel heat capacity and the temperature coefficient, $\gamma = |\alpha|/C$. Writing the prompt reactivity above critical as $\rho_0 - \beta$ and feedback $-\gamma E$, the power is

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{(\rho_0 - \beta) - \gamma E}{\Lambda}\, p. \tag{9.3}$$

Using $\mathrm{d}p/\mathrm{d}t = (\mathrm{d}p/\mathrm{d}E)\, p$ and integrating once,

$$p(E) = p_0 + \frac{1}{\Lambda}\left[(\rho_0 - \beta)E - \tfrac{1}{2}\gamma E^2\right]. \tag{9.4}$$

9.3 Peak power, energy, and temperature rise

The power peaks when $\mathrm{d}p/\mathrm{d}t = 0$, i.e. when the feedback exactly cancels the prompt reactivity, $\gamma E_{\text{peak}} = \rho_0 - \beta$, so the energy released at the peak is

$$\boxed{E_{\text{peak}} = \frac{\rho_0 - \beta}{\gamma}} \tag{9.5}$$

The excursion does not stop here—the power is maximal, not zero—but the feedback now drives the reactivity below prompt-critical and the power falls. The total energy released as the prompt spike completes is, by the symmetry of the parabola $p(E)$, twice the energy at the peak:

$$\boxed{E_{\text{total}} \approx \frac{2(\rho_0 - \beta)}{\gamma}} \tag{9.6}$$

and the peak power is

$$p_{\max} \approx \frac{(\rho_0 - \beta)^2}{2\gamma\Lambda}. \tag{9.7}$$

The corresponding adiabatic fuel temperature rise is $\Delta T_f = E_{\text{total}}/C = 2(\rho_0 - \beta)/(\gamma C)$. These formulae are the quantitative essence of self-limitation: the stronger the negative feedback γ, the smaller the energy release and temperature rise; the larger the prompt reactivity excess $\rho_0 - \beta$, the larger and sharper the spike.

Example 9.1 (A benign spike vs. a destructive one)**.** With prompt feedback strong enough that γ limits ΔT_f to a few hundred kelvin, an RIA produces a fast spike that the fuel survives—this is the designed-for behaviour validated in tests such as SPERT and BORAX. If instead the *net* prompt feedback is weak, near zero, or—catastrophically—positive, E_{total} grows without the parabolic turnover, the temperature rise is unbounded in the model, and in reality the energy deposition fragments the fuel, flashes the coolant, and disassembles the core. Chernobyl's excursion deposited enough energy in a fraction of a second to shatter the fuel and trigger a steam explosion, precisely because the prompt feedback in its configuration was not saving-ly negative.

9.4 Self-disassembly and the ultimate shutdown

In the most violent excursions a second, brutal feedback eventually intervenes: the core physically disassembles. Thermal expansion, fuel vaporisation, and pressure-driven dispersal increase neutron leakage and separate the fissile material, terminating the chain reaction—the Fuchs–Hansen "disassembly" mechanism that ended the historical criticality excursions (SL-1, the Godiva and SPERT destructive tests). Disassembly is a shutdown mechanism only in the sense that an explosion ends a chain reaction: it does so by destroying the reactor. The purpose of thermal stability and prompt negative feedback is to terminate the excursion by gentle means—temperature feedback—long before disassembly is reached.

9.5 Connection to the linear picture

The Nordheim–Fuchs excursion and the linear stability of Chapter 8 are two limits of the same coupled system. Linear analysis governs small perturbations about an operating point and asks whether they grow; the Nordheim–Fuchs analysis governs the large, fast, prompt-critical transient and asks how the nonlinear feedback caps it. A reactor that is linearly stable with strong negative feedback is also the one whose Nordheim–Fuchs excursions are mild. A reactor that is linearly unstable—positive feedback—has no benign nonlinear limit; its excursions terminate only by disassembly. The companion computations of Chapter 19 show both behaviours side by side.

Key point. In a prompt-critical excursion, prompt negative feedback caps the energy release at $E_{\text{total}} \approx 2(\rho_0 - \beta)/\gamma$ and the reactor survives. Without net negative prompt feedback there is no such cap, and the excursion ends only by destroying the core. Thermal stability is the difference between a spike and an explosion.

9.6 Exact solution of the Nordheim–Fuchs model

The Nordheim–Fuchs excursion admits a closed-form solution, which we now derive rigorously. It is one of the few exactly solvable nonlinear models in reactor dynamics, and its sech^2 pulse is exact, not approximate.

On the prompt timescale drop the precursors and write the prompt kinetics with adiabatic linear feedback,

$$\frac{\mathrm{d}p}{\mathrm{d}t} = a(t)\,p, \qquad a(t) = \frac{\rho_0 - \beta - \gamma E(t)}{\Lambda}, \qquad \frac{\mathrm{d}E}{\mathrm{d}t} = p, \tag{9.8}$$

where $E(t) = \int_0^t p\,\mathrm{d}t'$ is the released energy and $\gamma = |\alpha|/C > 0$ the feedback per unit energy.

Theorem 9.2 (Nordheim–Fuchs pulse)**.** *Let $a_0 = (\rho_0 - \beta)/\Lambda > 0$ (prompt supercritical) and $p(-\infty) = 0$. Then the solution of (9.8) is the symmetric pulse*

$$\boxed{p(t) = p_{\max}\operatorname{sech}^2\left(\frac{t - t_m}{2\tau}\right)}, \qquad p_{\max} = \frac{(\rho_0 - \beta)^2}{2\gamma\Lambda}, \quad \tau = \frac{\Lambda}{\rho_0 - \beta}, \tag{9.9}$$

with peak at the time t_m where $a(t_m) = 0$. The total released energy and the adiabatic temperature rise are

$$E_\infty = \int_{-\infty}^{\infty} p\,\mathrm{d}t = \frac{2(\rho_0 - \beta)}{\gamma}, \qquad \Delta T = \frac{E_\infty}{C} = \frac{2(\rho_0 - \beta)}{\gamma C}, \tag{9.10}$$

and the full width at half maximum is $\Delta t_{1/2} = 4\tau \operatorname{arccosh}\sqrt{2} \approx 3.53\,\tau$.

Proof. From (9.8), $a = \dot{p}/p = (\ln p)'$ and $\dot{a} = -(\gamma/\Lambda)\dot{E} = -(\gamma/\Lambda)p$. Hence $u = \ln p$ satisfies the Liouville equation $u'' = -(\gamma/\Lambda)e^u$. Multiplying by u' and integrating gives the first integral

$$\tfrac{1}{2}a^2 + \frac{\gamma}{\Lambda}p = K = \text{const.} \tag{9.11}$$

As $t \to -\infty$, $p \to 0$ and $a \to a_0$, so $K = \frac{1}{2}a_0^2$ and $a^2 = a_0^2 - 2(\gamma/\Lambda)p$. The peak ($\dot{p} = 0 \Rightarrow a = 0$) thus occurs at $p_{\max} = \Lambda a_0^2/(2\gamma) = (\rho_0 - \beta)^2/(2\gamma\Lambda)$. Substituting $a = \dot{p}/p$ into (9.11) yields a separable first-order ODE whose solution with $p(t_m) = p_{\max}$ is $p(t) = p_{\max}\operatorname{sech}^2[a_0(t - t_m)/2]$; identifying $\tau = 1/a_0$ gives the stated form. Finally $E_\infty = p_{\max}\int_{-\infty}^{\infty}\operatorname{sech}^2[(t - t_m)/2\tau]\,\mathrm{d}t = p_{\max} \cdot 4\tau = 2(\rho_0 - \beta)/\gamma$, and the FWHM follows from $\operatorname{sech}^2(z) = \frac{1}{2} \Leftrightarrow z = \operatorname{arccosh}\sqrt{2}$. □

The invariant (9.11) is the key structural fact: it is a conserved "energy" for the excursion, and its existence is exactly why a *negative* feedback ($\gamma > 0$) bounds the pulse, while $\gamma \leq 0$ destroys the invariant and the solution diverges. The released energy $E_\infty = 2(\rho_0 - \beta)/\gamma$ scales *inversely* with the feedback strength: a reactor with weak (or wrong- sign) prompt feedback deposits unbounded energy, the quantitative statement of the Chernobyl outcome.

Corollary 9.3 (No benign limit without negative feedback)**.** *If $\gamma \le 0$ the invariant* (9.11) *has no turning point with $p > 0$: $a(t)$ is nondecreasing and $p(t) \to \infty$ in finite or infinite time within the model. A bounded, self-terminating excursion exists if and only if the prompt feedback is strictly negative, $\gamma > 0$.*

Proof. The peak requires $a = 0$ with $p_{\max} = \Lambda a_0^2/(2\gamma) > 0$, impossible for $\gamma \le 0$. With $\gamma \le 0$, $\dot{a} = -(\gamma/\Lambda)p \ge 0$, so a never decreases from $a_0 > 0$ and $p = e^u$ grows without bound. □

Exercises

1. Derive $E_{\text{peak}} = (\rho_0 - \beta)/\gamma$ from the Nordheim–Fuchs model.
2. Show that the total energy release is approximately $2(\rho_0 - \beta)/\gamma$.
3. How does the peak power scale with the prompt reactivity excess and the feedback strength?
4. What physically terminates an excursion when prompt feedback is absent?
5. Relate the sign of the feedback in the Nordheim–Fuchs model to the linear stability of Chapter 8.

CHAPTER 10

Xenon Dynamics and Spatial Instability

Not all instabilities are fast. The slowest and most insidious arise from ^{135}Xe, a fission product with an enormous neutron-absorption cross-section whose concentration lags the power by hours. Xenon dynamics produce the post-shutdown "dead time" that constrains operation, and—in large cores—slow spatial power oscillations. Both played a direct role at Chernobyl: it was xenon poisoning, following an unplanned power drop, that lured the operators into withdrawing nearly all the control rods, stripping the reactor of the shutdown margin that might have saved it.

10.1 The iodine–xenon chain

^{135}Xe is produced in two ways: directly from fission (a small yield) and, dominantly, from the beta-decay of its precursor ^{135}I (half-life $\sim 6.6\,\mathrm{h}$), which is itself a fission product (yield $\gamma_I \approx 6.4\%$). ^{135}Xe is destroyed in two ways: by its own beta-decay (half-life $\sim 9.2\,\mathrm{h}$) and by neutron capture—"burnup"—into ^{136}Xe. Its thermal absorption cross-section is about 2.6×10^6 barn, among the largest known, so even trace amounts poison the reactor. The governing equations are

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \gamma_I \Sigma_f \phi - \lambda_I I, \tag{10.1}$$

$$\frac{\mathrm{d}X}{\mathrm{d}t} = \gamma_X \Sigma_f \phi + \lambda_I I - \lambda_X X - \sigma_X^a \phi X, \tag{10.2}$$

where I and X are the iodine and xenon concentrations, ϕ the flux, and σ_X^a the xenon absorption cross-section. The xenon contributes a negative reactivity $\rho_X \propto -\sigma_X^a X$.

10.2 Equilibrium xenon and its flux dependence

At steady flux the concentrations reach equilibrium,

$$I_\infty = \frac{\gamma_I \Sigma_f \phi}{\lambda_I}, \qquad X_\infty = \frac{(\gamma_I + \gamma_X)\Sigma_f \phi}{\lambda_X + \sigma_X^a \phi}. \tag{10.3}$$

At high flux, $\sigma_X^a \phi \gg \lambda_X$ and X_∞ saturates: burnup removes xenon as fast as it is made. The equilibrium xenon reactivity worth in a high-power thermal reactor is large, of order a few per cent Δk—several dollars—which must be compensated by control. The crucial nonlinearity is that production of xenon (via iodine) depends on the *past* flux, while burnup depends on the *present* flux; this time mismatch is the engine of every xenon transient.

10.3 The post-shutdown xenon peak and dead time

Suppose the reactor is shut down (or its power sharply reduced) from a high flux. Burnup of xenon ($\sigma_X^a \phi X$) stops immediately because ϕ collapses, but the stockpiled iodine keeps decaying into xenon for hours. The xenon concentration therefore *rises* after shutdown, peaking several hours later before the slower xenon decay finally wins and it falls. This transient buildup adds a large negative reactivity—the "xenon peak" or "iodine pit." If the reactor's excess reactivity is insufficient to override the peak, it cannot be restarted until the xenon decays: the **xenon dead time**, typically many hours. Operators must either restart promptly (before the peak) or wait it out.

Remark 10.1 (Chernobyl and the xenon trap). On the night of 25–26 April 1986, the Chernobyl test was delayed and the reactor power was held low and then dropped far below plan—down to about 30 MW thermal—for an extended period. Xenon built up and poisoned the core. To raise power against the xenon, operators withdrew control rods far beyond the permitted operating reactivity margin. The reactor was thus held critical only by a dangerously small number of inserted rods, with most absorbers removed—a configuration in which both the shutdown capability and the void coefficient were at their worst. Xenon dynamics did not cause the explosion, but they set the trap.

10.4 Spatial xenon oscillations

In a physically large core—and the RBMK was very large, about 12 m in diameter—the point approximation fails for slow phenomena, and xenon can drive *spatial* oscillations even at constant total power. The mechanism is a delayed, out-of-phase feedback between flux and xenon:

1. A perturbation raises the flux in one region (say, the top of the core) and lowers it in another (the bottom).
2. In the high-flux region, xenon *burns out* faster than it is produced, so its concentration falls, which *raises* the local reactivity and pushes the flux even higher.
3. Meanwhile the extra iodine produced earlier in that region decays into xenon hours later, eventually *raising* the xenon and suppressing the flux there—just as the other region, now iodine-rich, begins to flare.
4. The high-flux and low-flux regions thus trade places with a period of roughly $15-30$ h.

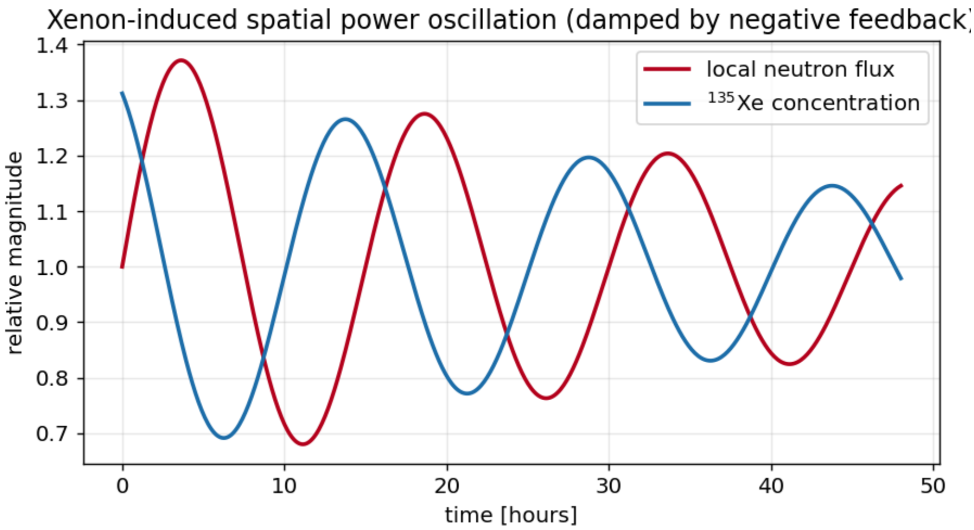

Figure 10.1: Xenon-induced spatial power oscillation. The local flux and the ^{135}Xe concentration oscillate out of phase with a period of order a day. A sufficiently negative power (temperature) coefficient damps the oscillation; without it, the amplitude can grow.

Whether these oscillations grow or decay depends on the power coefficient. A strong *negative* temperature/power coefficient damps them: as a region's power rises, its temperature rises and removes reactivity, opposing the xenon-led growth. A weak or positive coefficient leaves the oscillation undamped or growing. Large reactors counter spatial xenon instability with negative feedback and with spatial control (segmented control rods, flux mapping, and active power-shaping). The RBMK required constant operator attention to its axial and radial power distribution precisely because of its size and its marginal feedback; the night of the accident, the flux shape was badly distorted (axially peaked), worsening the local void and power coefficients beyond what core-average figures suggested.

10.5 Implications for stability and operation

Xenon dynamics teach two lessons relevant to thermal stability. First, the reactor's state depends on its *history*: the same power level can correspond to very different xenon (and hence reactivity and rod) configurations depending on what the power did over the preceding day. Operating procedures must account for this memory. Second, spatial effects mean that a core can be "stable on average" yet locally unstable; core-average coefficients can mislead. Both lessons were violated at Chernobyl, where a history of power manoeuvring produced a xenon-poisoned, rod-depleted, axially-distorted core whose local stability was far worse than any single average number would suggest.

Key point. Xenon-135 introduces hours-long memory into reactor dynamics: a post-shutdown poisoning peak that can preclude restart, and—in large cores—slow spatial power oscillations damped only by negative feedback. At Chernobyl, xenon poisoning drove the operators to strip out the control rods, creating the low-margin, distorted-flux state in which the positive void coefficient could trigger catastrophe.

10.6 Linear stability of the iodine–xenon equilibrium

We now analyse xenon dynamics rigorously, first at a point (temporal stability) and then for a spatial mode (the oscillation).

Temporal stability at fixed flux. Linearise the iodine–xenon equations about the equilibrium (I_∞, X_∞) at constant flux ϕ_0. With state $(\delta I, \delta X)$ the Jacobian is

$$\mathbf{J} = \begin{pmatrix} -\lambda_I & 0 \\ \lambda_I & -(\lambda_X + \sigma_X \phi_0) \end{pmatrix}, \tag{10.4}$$

which is lower-triangular with eigenvalues $-\lambda_I < 0$ and $-(\lambda_X + \sigma_X\phi_0) < 0$. Hence at fixed flux the point equilibrium is *unconditionally asymptotically stable*: a purely temporal xenon transient (the post-shutdown peak) decays. Instability requires the flux to respond to xenon, i.e. spatial coupling.

Spatial mode and the oscillation. Consider the first antisymmetric flux mode with amplitude $\epsilon(t)$ (a "tilt" between two halves of the core) coupled to the differential xenon $\eta(t)$ and differential iodine $\zeta(t)$ between the halves. Retaining the xenon reactivity feedback on the tilt and a prompt power (temperature) coefficient α_P, the linearised modal system takes the form

$$\dot{\epsilon} = g\,(c_X\,\eta + \alpha_P\,\epsilon), \quad \dot{\zeta} = \gamma_I \Sigma_f \phi_0\,\epsilon - \lambda_I \zeta, \quad \dot{\eta} = \lambda_I \zeta - (\lambda_X + \sigma_X\phi_0)\eta - \sigma_X \phi_0 X_\infty\,\epsilon, \tag{10.5}$$

where $g > 0$ is the modal flux gain and $c_X > 0$ encodes the destabilising xenon-burnout coupling. The Jacobian of (10.5) is 3×3 with characteristic polynomial $s^3 + b_1 s^2 + b_2 s + b_3$ whose coefficients depend on α_P through $b_1 \ni -g\alpha_P$.

Proposition 10.2 (Feedback threshold for spatial xenon stability)**.** *There is a critical coefficient* $\alpha_{\text{crit}}(\phi_0) > 0$*, increasing in the flux* ϕ_0 *and in the core size, such that the spatial xenon mode* (10.5) *is asymptotically stable if and only if*

$$\alpha_P < -\,\alpha_{\text{crit}}(\phi_0). \tag{10.6}$$

For $\alpha_P > -\alpha_{\text{crit}}$ *a complex-conjugate pair of eigenvalues crosses into the right half-plane (a Hopf bifurcation), producing a growing oscillation whose angular frequency at onset is* $\omega_{\text{osc}} \approx \sqrt{b_2}$*, corresponding to a period of order* 15–30 h *for typical thermal-reactor parameters.*

Sketch. Apply the Routh–Hurwitz criterion (Theorem 8.2) to $s^3+b_1s^2+b_2s+b_3$. The destabilising xenon coupling enters b_1, b_2, b_3 with a sign opposite to α_P; a more negative α_P increases b_1 and the product $b_1b_2 - b_3$. The condition $b_1b_2 - b_3 > 0$ defines a threshold α_{crit} such that the boundary $b_1b_2 - b_3 = 0$ is the Hopf locus, where a conjugate pair $s = \pm i\omega_{\text{osc}}$ with $\omega_{\text{osc}}^2 = b_2$ sits on the imaginary axis. The dependence of α_{crit} on ϕ_0 and core size follows from $c_X, g \propto \phi_0$ and the larger spatial gain of bigger cores. □

Proposition 10.2 formalises the qualitative claim of Figure 10.1: a sufficiently strong negative power coefficient damps the spatial xenon oscillation, while a weak or positive one permits a growing, day-scale oscillation—most easily in a large core such as the RBMK.

Exercises

1. Write the iodine–xenon equations and find the equilibrium xenon concentration.
2. Explain why xenon *builds up* after a shutdown from high power.
3. Describe the mechanism of spatial xenon oscillations and the role of negative feedback in damping them.
4. Explain how xenon poisoning contributed to the loss of operating reactivity margin at Chernobyl.
5. Why are large cores more susceptible to spatial xenon instability than small ones?

CHAPTER 11

Reactor Control, Period, and Protection Systems

Stability theory tells us how a reactor responds to disturbances; control and protection systems are how engineers ensure the reactor stays within the stable, safe envelope and is shut down if it strays. This chapter connects the dynamics of the previous chapters to the practical machinery of reactivity control, period measurement, and emergency protection—and shows precisely which of these safeguards were defeated or flawed at Chernobyl.

11.1 Means of reactivity control

Reactivity is adjusted by several means with very different speeds and purposes. **Control rods**—neutron absorbers (boron carbide, hafnium, silver-indium-cadmium) —provide fast, localised, and reliable control; they are the primary means of prompt shutdown. **Soluble chemical shim**—boric acid dissolved in the coolant of a PWR—provides slow, spatially uniform control of the large, slow reactivity changes of fuel burnup and xenon. **Burnable poisons**—fixed absorbers that deplete with irradiation—offset the excess reactivity of fresh fuel. The art of control is to match each tool to the timescale of the reactivity change it must counter.

11.2 Rod worth and the integral/differential curves

The reactivity inserted by a control rod depends nonlinearly on its position because it is proportional to the local flux-squared (the rod removes neutrons in proportion to the flux, and is most effective where the flux is highest—near the core centre). The *differential rod worth* $\mathrm{d}\rho/\mathrm{d}z$ is therefore largest near mid-core and small at the ends, and the *integral worth* $\rho(z)$ is the characteristic S-shaped curve. Two consequences matter for stability. First, a rod moving through the high-worth central region inserts reactivity rapidly—a hazard if withdrawn carelessly. Second, the axial flux shape determines where a rod's worth lies; in the RBMK, with the flux skewed to the bottom of the core, the bottom-entering graphite displacers acted exactly where the flux—and hence the harmful positive insertion—was concentrated.

11.3 The reactor period and its measurement

The reactor period $T = 1/s$ (Chapter 3) is the most direct measure of how fast power is changing. Operators and protection systems monitor the period with *period meters* (or the equivalent "startup rate" in decades per minute). A short period is an alarm: protection systems trip the reactor if the period falls below a set limit (e.g. a few seconds), because a short period signals approach to dangerous reactivity. The inhour equation links the period directly to reactivity, so a period-meter is, in effect, a real-time reactivity gauge. Period-based trips are a first line of defence against reactivity excursions—though against a prompt-critical insertion, the period collapses faster than any trip can act, which is why *inherent* prompt feedback, not protection, is the true safeguard at that extreme.

11.4 Protection systems and defence in depth

The reactor protection system (RPS) monitors many variables—neutron flux, rate of change (period), coolant temperature, pressure, flow, and water level—and initiates an automatic scram when any exceeds a safe limit. The design philosophy is *defence in depth*: multiple independent, redundant, and diverse barriers, so that no single failure (and ideally no plausible combination) leads to harm. A protection system is engineered to be *fail-safe*: loss of power or signal drives the rods in, not out.

Remark 11.1 (Defeated protections at Chernobyl). The Chernobyl test required, and the operators implemented, the deliberate *disabling* of several automatic protections—including the trip on turbine stop and certain emergency-cooling and flux signals—so that the test would not be interrupted. Defeating protections is sometimes necessary for testing, but it removes the very defence-in-depth barriers that exist to catch the unexpected. Combined with operation in an unstable regime, it left the reactor with neither inherent stability nor engineered protection. The single remaining safeguard—the AZ-5 scram—was itself flawed by the positive scram effect. Every layer of defence had been removed or inverted.

11.5 The speed problem

A recurring theme is the mismatch between the timescales of protection and of a prompt excursion. Mechanical scram—unlatching rods and letting them fall or drive in—takes on the order of one to a few seconds in a fast modern system, and was about 18 s for full insertion in the RBMK. A delayed-critical transient ($\rho < \beta$) evolves over seconds and can be caught. A prompt-critical transient ($\rho > \beta$) evolves over milliseconds and cannot. This is why the design objective is to make prompt criticality *unreachable* by normal means and to ensure that the prompt feedback is negative: protection systems guard the delayed-critical regime, but only physics guards the prompt regime.

Key point. Control and protection systems keep a reactor within its stable envelope and shut it down if it strays—but they act on the delayed-neutron timescale of seconds. Against a prompt-critical excursion, only inherent prompt negative feedback is fast enough. At Chernobyl the protections were disabled, the one remaining scram was flawed, and the inherent feedback was positive: defence in depth had been hollowed out at every layer.

Exercises

1. Distinguish control rods, soluble shim, and burnable poisons by speed and purpose.

2. Sketch the integral and differential rod-worth curves and explain their shape.
3. Why does a short reactor period trigger a protective scram, and why is this useless against a prompt-critical insertion?
4. Define defence in depth and explain how it was hollowed out at Chernobyl.
5. Explain the fail-safe principle for a reactor protection system.

CHAPTER 12

Measuring Reactivity and Feedback Coefficients

A reactor's stability rests on the signs and magnitudes of its feedback coefficients. These are not merely calculated; they are *measured*, both during commissioning and periodically through life. This chapter surveys the experimental methods, which double as a practical illustration of the kinetics theory and underline a sobering point: the RBMK's dangerous coefficients were, in principle, measurable and partly known.

12.1 Static and dynamic reactivity measurement

Rod-drop and source-jerk methods infer reactivity from the transient following a sudden change. In a rod-drop, a calibrated rod is dropped and the prompt power decrease is fitted to the point-kinetics solution to extract the inserted negative reactivity. **Rod-oscillator methods** insert a sinusoidal reactivity and measure the power response; the amplitude and phase give the zero-power transfer function $Z(j\omega)$ of Chapter 8 directly, mapping the reactor's dynamic response across frequency. **Inverse kinetics**—solving the point-kinetics equations "backwards" for $\rho(t)$ given a measured power trace—is the basis of the modern *reactivity meter*, a real-time instrument used during physics testing.

12.2 Period measurement and the inhour calibration

The simplest reactivity measurement is the *positive-period* method: insert a small known reactivity, measure the resulting stable period, and confirm it against the inhour equation (3.5). This both measures reactivity and validates the kinetics data. The agreement between measured periods and the inhour prediction (Figure 3.1) is, in effect, a routine check that the reactor's delayed-neutron parameters and generation time are as expected.

12.3 Measuring temperature and void coefficients

The **isothermal temperature coefficient** is measured at low power by slowly heating the whole coolant/moderator system (using pump heat, with the reactor just critical) and recording the compensating rod motion: the rod worth needed to hold criticality per degree is the coefficient. The **moderator temperature coefficient (MTC)** and **Doppler (fuel) coefficient** can be separated by exploiting their different timescales—the fuel responds promptly, the moderator with the coolant transport delay—using power-oscillation or noise techniques. The **void coefficient** is inferred from controlled changes in boiling or moderator level, or in test reactors by deliberately introducing voids. The **power coefficient** is measured directly by recording the reactivity (via the reactivity meter) required to change power.

12.4 Noise analysis: listening to the reactor

A powerful, non-invasive technique is *reactor noise analysis*. The neutron flux fluctuates around its mean due to the statistical nature of fission and the reactor's dynamic response; the spectrum of these fluctuations encodes the transfer function and hence the feedback. Measuring the *decay ratio* (Chapter 8) from the autocorrelation of small flux oscillations is the standard method for monitoring BWR stability margins online—a direct, continuous read on how close the reactor is to the oscillatory stability boundary.

Remark 12.1 (The knowable danger). The RBMK's positive void coefficient and the positive scram effect were, in principle, measurable quantities, and the scram anomaly had in fact been observed at the Ignalina plant in 1983. The tragedy is not that the danger was unmeasurable but that the measurements and observations were not translated into design fixes, clear operating limits, and operator understanding. Measurement is only the first step; acting on it is safety.

Key point. Feedback coefficients are measured by rod-drop, oscillator, inverse kinetics, isothermal heating, and noise methods, and validated against the inhour equation. The RBMK's destabilising coefficients were measurable and partly observed; the failure was organisational—not converting knowledge into design changes and enforced limits.

Exercises

1. Describe the rod-drop method for measuring reactivity.
2. How does a rod oscillator measure the zero-power transfer function?
3. Outline how the isothermal temperature coefficient is measured.
4. Explain how reactor noise analysis monitors BWR stability margins.
5. Discuss why the RBMK danger was knowable yet not acted upon.

Part IV

Historical Lessons

CHAPTER 13

A History of Stability-Related Incidents

Chernobyl was not the first time reactor dynamics asserted themselves. A series of earlier events, some deliberate experiments and some accidents, mapped out the behaviour of reactors near the edge of stability and shaped the safety principles that the RBMK, tragically, did not fully embody. This chapter surveys the most instructive cases as context for the Chernobyl analysis to follow.

13.1 Deliberate destructive tests: BORAX and SPERT

In the 1950s and 1960s the United States conducted a remarkable series of *intentional* excursion experiments to learn how reactors behave when pushed past prompt criticality. The BORAX and SPERT (Special Power Excursion Reactor Test) programmes inserted large step reactivities into water-moderated reactors and measured the response. The central finding validated the Nordheim–Fuchs picture: in an under-moderated water reactor, the prompt negative void and Doppler feedback produced a self-limiting power spike. As the water heated and boiled, the negative void coefficient sharply reduced reactivity and terminated the excursion—often before any mechanical action. The BORAX-I reactor was eventually destroyed in a deliberately oversized excursion, confirming that even when the energy release fragments the core, the feedback that ends the transient is the machine's own physics. These tests are the empirical foundation for the claim that a negative-void, negative-Doppler reactor is inherently self-limiting.

13.2 SL-1 (1961): a prompt-critical fatal excursion

The SL-1 (Stationary Low-Power Reactor No. 1) accident in Idaho on 3 January 1961 was the first and only fatal reactor accident in U.S. history. A single central control rod was manually withdrawn far beyond its normal range—by hand, during maintenance—inserting a large prompt-critical reactivity in an instant. The power rose to perhaps 20 GW within milliseconds; the water flashed to steam and the resulting pressure surge (a water-hammer) lifted the entire reactor vessel several feet, killing the three operators. SL-1 demonstrated, lethally, two themes that recur at Chernobyl: a single excessive reactivity insertion can take a reactor prompt-critical faster than anyone can react, and the destructive energy is delivered not directly by neutrons but by the resulting steam explosion.

13.3 NRX (1952) and the Windscale fire (1957)

Canada's NRX reactor suffered a partial meltdown in 1952 after a combination of operator error and control-rod malfunction led to a power excursion with inadequate shutdown; it became an early lesson in the importance of reliable, fast, and unambiguous shutdown systems. The 1957 Windscale fire in the United Kingdom, though primarily a graphite-fire and Wigner-energy event rather than a prompt excursion, underscored the hazards of graphite-moderated, gas-cooled reactors and of stored energy in graphite—issues that resonate with the RBMK's graphite moderator.

13.4 Fermi-1 (1966): partial meltdown in a fast reactor

The Enrico Fermi-1 sodium-cooled fast breeder reactor suffered a partial fuel meltdown in 1966 when a loose flow-blocking plate starved several fuel assemblies of coolant. Although triggered by a flow blockage rather than a reactivity event, Fermi-1 highlighted the tight coupling between thermal-hydraulics and core integrity, and the consequences of local cooling failure—an enduring theme in thermal safety.

13.5 Boiling-water reactor instability events

Boiling-water reactors carry a designed-in stability concern because their void feedback is transported by the coolant and therefore phase-lagged (Chapter 8). Several plants experienced unintended power oscillations in low-flow, high-power conditions. The most cited is the **LaSalle Unit 2 event of 1988**, in which a loss of feedwater heating and a trip of both recirculation pumps left the reactor at high power and low (natural circulation) flow—exactly the regime of weakest stability margin. The reactor entered a growing power oscillation (an increasing decay ratio) before operators manually scrammed it. No damage resulted, but the event prompted industry-wide changes: stability exclusion regions on the power–flow map, oscillation detection systems, and revised procedures. LaSalle is the clearest modern illustration of a *linearly* unstable operating point in a commercial reactor—and, unlike Chernobyl, of a system caught and shut down before harm.

13.6 Three Mile Island (1979): the thermal aftermath

The Three Mile Island Unit 2 accident was not a reactivity-stability event—the reactor scrammed correctly and promptly. It belongs in this history for the *other* half of thermal safety: decay heat. A stuck-open relief valve and misread instruments led operators to throttle emergency cooling, and decay heat (Chapter 4) slowly uncovered and melted much of the core over hours. TMI is the canonical reminder that stopping fission does not stop the heat, and that human factors and instrumentation are part of the safety system. Its lessons about operator training, control-room design, and instrumentation directly informed the international response to Chernobyl seven years later.

13.7 What the history teaches

Read together, these events sketch a coherent set of principles:

- A single large, fast reactivity insertion can take a reactor prompt-critical faster than any intervention (SL-1).
- In a reactor with prompt *negative* feedback, excursions are self-limiting (BORAX, SPERT)—this is the design goal.
- Phase-lagged feedback can make a nominally negative-feedback reactor oscillate (LaSalle); margins and monitoring are required.
- Decay heat makes loss of cooling dangerous even after shutdown (TMI, Fermi-1).
- Shutdown systems must be fast, reliable, and unambiguous (NRX).

The RBMK violated or weakened several of these at once: it had a positive feedback mechanism (void), it was operated into its least stable regime, its shutdown system had a fatal flaw, and its operators acted on an incomplete understanding of the machine's physics. The next chapters examine how.

Exercises

1. What did the BORAX and SPERT tests demonstrate about under-moderated water reactors?
2. Summarise the SL-1 accident and the two general lessons it shares with Chernobyl.
3. Explain why the LaSalle-2 event is an example of a linearly unstable operating point.
4. Why is Three Mile Island classified here as a decay-heat rather than a stability accident?
5. Extract five general safety principles from the historical record.

CHAPTER 14

The RBMK-1000 Reactor

To understand Chernobyl one must understand the machine. The RBMK-1000 (*reaktor bolshoy moshchnosti kanalnyy*—high-power channel-type reactor) was a uniquely Soviet design whose distinctive features—graphite moderation, water cooling in individual pressure tubes, on-load refuelling, and a very large core—combined to give it the destabilising characteristics that the accident exposed. This chapter dissects the design with particular attention to the features that bear on thermal stability: the positive void coefficient and the flawed control-rod design.

14.1 Design philosophy and layout

The RBMK was conceived to produce both electricity and, in its lineage, plutonium, and to use natural or low-enriched uranium with on-load refuelling and domestically available technology. It dispensed with the massive single pressure vessel of a Western PWR. Instead, the core was an enormous stack of graphite blocks—about 12 m in diameter and 7 m tall, weighing some 1700 tonnes—penetrated by roughly 1660 vertical **pressure tubes** (fuel channels). Each channel held fuel assemblies of slightly enriched ($\sim 2\%$) UO_2 and carried boiling light water as coolant. The water boiled in the channels (like a BWR), and the steam was separated and sent to the turbines.

The decisive architectural fact is the *separation of moderator and coolant*. In a PWR or BWR the water is both moderator and coolant, so losing water loses moderation. In the RBMK the **graphite** did the moderating, and the water was mainly coolant—and, because water absorbs thermal neutrons, also a parasitic absorber. This separation is the root of the positive void coefficient.

14.2 The positive void coefficient

Consider what happens when the cooling water in an RBMK channel boils more vigorously, increasing the steam void fraction. Two effects compete, as in any water reactor (Chapter 6): loss of moderation (tends to reduce reactivity) and loss of neutron absorption (tends to increase it). In the RBMK the graphite continues to moderate regardless of the water, so the loss-of-moderation effect is weak; what dominates is the removal of the water *absorber*. Boiling the water therefore *adds* reactivity:

$$\alpha_v = \frac{\partial \rho}{\partial \bar{v}} > 0, \tag{14.1}$$

strongly positive. The magnitude depended on the core state—fuel burnup, enrichment, and especially the number of control rods inserted and the operating reactivity margin. In the configuration of the accident, the void coefficient was large enough that the total power coefficient at low power was *positive*: a rise in power boiled more water, which raised reactivity, which raised power—a self-amplifying loop with no prompt negative effect strong enough to counter it.

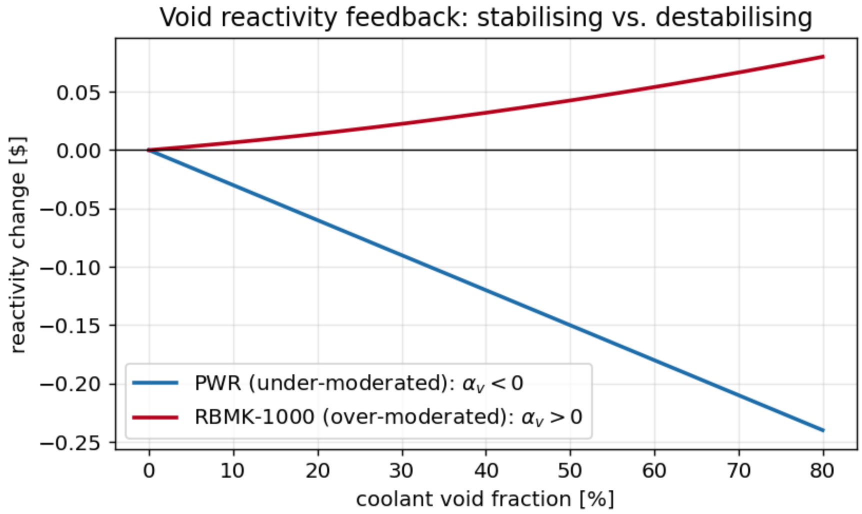

Figure 14.1: The RBMK's over-moderated lattice gives a positive void coefficient (rising curve): boiling adds reactivity, in contrast to the negative void coefficient of an under-moderated PWR. This is the central design flaw behind Chernobyl.

14.3 The graphite moderator and the positive feedback chain

The large graphite mass gave the RBMK a long prompt neutron lifetime and a great deal of thermal inertia, which in some respects made the reactor sluggish and seemingly forgiving. But the same graphite is what decoupled moderation from the coolant and produced the positive void coefficient. Graphite also stores energy (Wigner energy) and burns in air at high temperature—both relevant to the accident's aftermath, when the exposed graphite caught fire and lofted fission products for days. The design thus combined a destabilising neutronic characteristic with a flammable moderator, a doubly unfortunate pairing.

14.4 The fatal control-rod design

The RBMK's most notorious flaw lay in its control rods. Each absorber rod was fitted, on its lower end, with a **graphite "displacer" (follower)** about 4.5 m long, separated from the boron-carbide absorber by a short column of water. The purpose of the graphite tip was to keep water (an absorber) out of the channel below a withdrawn rod, improving efficiency. The consequence was lethal.

When a rod was fully withdrawn, its graphite displacer sat in the central region of the core and the lower portion of the channel was filled with water. As the rod began to insert from the fully-withdrawn position, the leading element to enter the core's lower region was not the boron absorber but the *graphite displacer*, which *displaced water* (an absorber) with *graphite* (a moderator). The immediate, local effect of beginning to insert the rod was therefore to *add* positive reactivity in the lower core—before the absorber section arrived to add the intended negative reactivity. This is the **positive scram effect**: pressing the emergency shutdown button could, for the first few seconds and especially with an axially bottom-peaked flux, momentarily *increase* reactivity.

Remark 14.1 (A known flaw, unaddressed). The positive scram effect was not unknown. A similar control-rod-insertion power spike had been observed at the Ignalina RBMK in 1983, and the phenomenon was understood in principle. Yet the design was not corrected and operators were not adequately warned. INSAG-7 later noted that there was a widespread belief that the conditions under which the positive scram effect would matter "would never occur"—conditions that, in the event, "appeared in almost every detail" at Chernobyl.

14.5 The operating reactivity margin

Because the RBMK relied on a substantial number of inserted control rods to shape its flux and maintain controllability, operating rules specified a minimum **operating reactivity margin (ORM)**—expressed as an equivalent number of fully-inserted control rods (nominally about 30, with a hard floor of 15). The ORM was, in effect, the reactor's shutdown reserve and its guarantee of controllability and acceptable feedback characteristics. Operating below the minimum ORM was prohibited because it left too few absorbers to shut the reactor down quickly and—crucially—worsened the void coefficient and the axial power shape. On the night of the accident the ORM was driven far below the limit, to the equivalent of perhaps six to eight rods, in the struggle against xenon poisoning.

14.6 The instrumentation and protection gaps

The RBMK's protection systems and instrumentation compounded the physical flaws. The emergency shutdown was slow—full rod insertion took on the order of 18 s, an eternity for a prompt excursion. Key automatic protection trips had been disabled to permit the test. The control system did not enforce the ORM limit automatically. And the operators' training and documentation did not convey the true severity of the low-power, low-ORM regime. The machine's safety thus depended on operators avoiding a state whose danger the machine neither prevented nor clearly signalled.

Key point. The RBMK-1000 combined a positive void coefficient (from graphite moderation decoupled from water cooling), a control-rod design that added reactivity in the first seconds of a scram (the positive scram effect), a large core prone to spatial flux distortion, a slow shutdown system, and weak enforcement of the operating reactivity margin. Each flaw was survivable alone; in combination, at low power with depleted margin, they made the reactor a prompt-excursion accident waiting for a trigger.

Exercises

1. Explain how separating moderator (graphite) from coolant (water) produces a positive void coefficient.
2. Describe the positive scram effect and the role of the graphite displacers.

3. What is the operating reactivity margin, and why does operating below it worsen stability?
4. List the RBMK design features that each contributed to the accident.
5. Why was the $\sim 18\,\mathrm{s}$ shutdown time a serious weakness?

CHAPTER 15

The Chernobyl Accident

On 26 April 1986, at 01:23:40 local time, an operator at Chernobyl Unit 4 pressed the AZ-5 emergency shutdown button. Within seconds the reactor power surged to many times its rated value, two explosions destroyed the core and building, and the worst nuclear accident in history began. This chapter reconstructs the sequence with the physics of the preceding chapters in hand. The aim is not to assign blame but to show the equations playing out in a real machine.

15.1 The test and its purpose

The accident occurred during a *safety test*, ironically intended to improve the plant's response to a station blackout. The test sought to verify that, after a loss of off-site power, the rotational inertia of a coasting-down turbine generator could power the main coolant pumps for the brief interval before the diesel generators reached full load. It was an electrical-engineering test of turbine coastdown, to be run at reduced reactor power. Crucially, conducting it required putting the reactor into a low-power state and disabling several protective systems—taking the machine toward the corner of its operating envelope where, as Chapter 14 showed, it was least stable.

15.2 The descent into the unstable regime

The test was planned for the daytime crew but was repeatedly delayed; the grid dispatcher requested that Unit 4 keep supplying power through the evening. The power reduction was thus stretched over many hours, during which two destabilising processes unfolded.

Power reduction and xenon buildup. The reactor was held at reduced power and then, around 00:28, its thermal power fell far below the test plan—to roughly 30 MW thermal, a small fraction of the 3200 MW rating. Accounts differ on whether this was an operator control error or an instrumentation/control fault; either way, the low power had a fateful consequence. During the preceding hours at higher power the core had accumulated ^{135}I; now, at very low flux, xenon burnup ceased while iodine kept decaying into xenon (Chapter 10). The core became heavily **xenon-poisoned**, and the power could not easily be raised.

Rod withdrawal and loss of margin. To fight the xenon and bring the power back up for the test, operators withdrew control rods—many of them. By the time power was coaxed to about 200 MW thermal (still well below the planned $700 - 1000$ MW), the **operating reactivity margin** had fallen far below its minimum, to the equivalent of only a handful of rods. The reactor was now held critical by very few absorbers, with most withdrawn—the precise state in which the void coefficient was largest, the axial flux shape distorted, and the shutdown capability hollowed out.

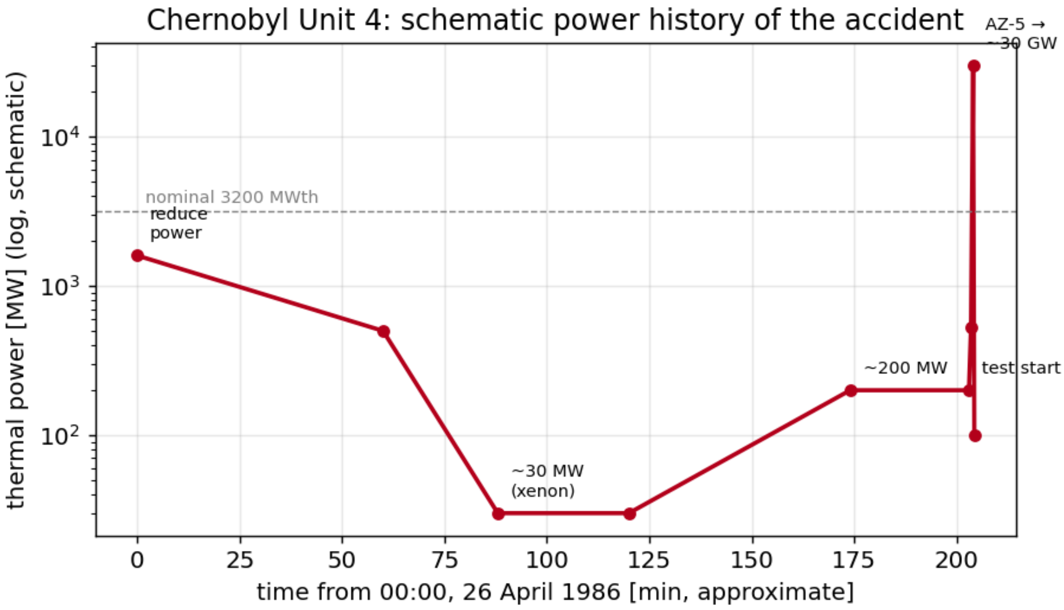

Figure 15.1: Schematic thermal-power history of Chernobyl Unit 4 through the night of the accident (illustrative, log scale). Power was reduced for the test, collapsed to ~ 30 MW with xenon poisoning, was partially recovered to ~ 200 MW by withdrawing rods, and then surged to tens of gigawatts in seconds after AZ-5 was pressed.

15.3 Starting the test

With the reactor at about 200 MW, the crew proceeded with the test rather than aborting—a low power at which the RBMK's instability was acute. To run the test they configured the plant in ways that further reduced safety margins: additional main coolant pumps were running (increasing core flow, reducing boiling and thus the steam void fraction, but pushing pumps toward cavitation), and several automatic protection signals—including those that would have scrammed the reactor on low water level or on the trip of the turbine—were blocked so that the test would not be interrupted.

At 01:23:04 the turbine steam supply was closed to begin the coastdown. As the main pumps driven by the coasting turbine slowed, core coolant flow decreased. Less flow meant more boiling in the channels; more boiling meant more steam *voids*; and because the void coefficient was strongly positive, the voids added reactivity. Power began to climb. In a negative-void reactor this would have been self-correcting; in the RBMK it was self-amplifying. The reactor was now sliding up the positive-feedback ramp.

15.4 AZ-5 and the power surge

At 01:23:40, with power rising, an operator pressed **AZ-5**, the emergency scram, to insert all control and protection rods. What should have shut the reactor down did the opposite—at first. As the rods began to descend from their nearly fully-withdrawn positions, the *graphite displacers* entered the lower core first, displacing water (an absorber) and **adding** positive reactivity in the bottom of the core (the positive scram effect of Chapter 14). Because the flux was already distorted toward the bottom of the core, this local positive insertion struck exactly where it would do the most harm.

The added reactivity pushed the reactor through prompt criticality. Now the physics of Chapters 3 and 9 took over on the millisecond timescale. With $\rho > \beta$, the power grew as $\exp[(\rho - \beta)t/\Lambda]$; the positive void feedback, far from arresting the rise, reinforced it as the surging power flashed the remaining water to steam; and there was no sufficient prompt negative feedback to cap the excursion. The power spiked to an estimated tens of gigawatts—perhaps a hundred times the rated power—in a few seconds.

15.5 The explosions

The energy deposited in this prompt excursion was enormous and almost instantaneous. The fuel overheated and fragmented; the intense heat flashed the cooling water to steam, and the pressure surge ruptured fuel channels. The first explosion was a **steam explosion**—the violent expansion of suddenly generated steam—which blew off the 1000-tonne upper biological shield ("Elena") and tore open the core. A second explosion followed seconds later; its exact nature is debated (a further steam/thermal explosion, possibly with a contribution from hydrogen generated by steam reacting with hot zirconium and graphite, and a brief renewed criticality in the disrupted core). The explosions demolished the reactor hall, ejected fuel and graphite, and exposed the core to the atmosphere.

15.6 The fire and the release

With the core breached and the graphite moderator exposed to air at high temperature, the **graphite caught fire**. The fire burned for days, lofting fission products—^{131}I, ^{137}Cs, ^{134}Cs, and others—high into the atmosphere, where winds carried them across the western Soviet Union and Europe. The decay heat (Chapter 4) in the wrecked fuel sustained high temperatures and continued to drive releases. Emergency crews fought the fire and later dropped boron, sand, clay, and lead from helicopters in an attempt to smother the core and absorb neutrons. The release continued for about ten days before subsiding.

15.7 The sequence in one paragraph

Stripped to essentials: a delayed, low-power test poisoned the core with xenon; fighting the xenon stripped out the control rods, leaving the reactor in its most unstable configuration with almost no shutdown margin; reducing coolant flow for the test boiled more water, and the positive void coefficient drove power upward; the emergency scram, through the graphite-tipped rods, added reactivity instead of removing it and pushed the reactor prompt-critical; the positive void feedback amplified the prompt excursion; and the resulting energy surge flashed the coolant and exploded the core. Every step is a chapter of this book made flesh. The next chapter analyses why, and how the understanding evolved.

Exercises

1. Reconstruct the chain of events from the power drop to the explosions in five steps.
2. Explain why reducing coolant flow for the test raised reactor power.
3. Why did pressing AZ-5 initially increase reactivity?
4. Estimate, qualitatively, why the excursion was over in seconds using the prompt timescale.
5. Distinguish the roles of the steam explosion and the graphite fire in the consequences.

CHAPTER 16

Chernobyl: Root-Cause Analysis

Why did Chernobyl happen, and what did the world conclude? The answer evolved significantly between 1986 and 1992, and the evolution is itself instructive: it is a case study in how the apportionment of blame between “human error” and “design flaw” depends on how deeply one understands the machine. This chapter analyses the accident through the feedback theory of Parts II and III and traces the shift from INSAG-1 to INSAG-7.

16.1 The accident as a feedback failure

In the language of this book, Chernobyl was the failure of a closed-loop control system whose loop gain became positive. Recall the stability principle: a reactor is stable when rising temperature/power removes reactivity faster than it is added. At Chernobyl, in its low-power, low-ORM, distorted-flux state, three feedback elements all had the wrong sign or were too weak:

- The **void coefficient** was strongly positive: rising power boiled water, which added reactivity (Chapters 6, 14).
- The **prompt Doppler feedback**, though negative, was too weak at low power to overcome the void term (the fuel-temperature swings at low power are small).
- The **scram**, intended as the ultimate negative reactivity, was momentarily *positive* because of the graphite displacers (the positive scram effect).

With the dominant feedbacks positive, the Nyquist locus of Chapter 8 enclosed the critical point, and the Nordheim–Fuchs balance of Chapter 9 had no benign turnover: there was no net negative prompt feedback to cap the excursion. The accident is, in this sense, exactly what the stability theory predicts for a machine with these coefficients driven into this regime. It was not an inexplicable catastrophe; it was the equations asserting themselves.

16.2 Contributing causes: a layered failure

No single fault caused the accident; it was a chain in which each link mattered.

Design causes. The positive void coefficient, the positive scram effect of the graphite-tipped rods, the slow ($\sim 18\,\mathrm{s}$) shutdown, the large core prone to spatial flux distortion, and the lack of automatic enforcement of the ORM—all were properties of the machine.

Operational causes. The test was conducted at far too low a power; the ORM was driven far below its minimum; protective trips were disabled; and the crew proceeded despite the reactor being in an unfamiliar and unstable state.

Organisational and cultural causes. The positive scram effect was known from the 1983

Ignalina event but neither corrected nor communicated; operating documentation understated the danger of the low-ORM regime; the test procedure was inadequate and inadequately reviewed; and a safety culture that treated such margins as bureaucratic rather than physical allowed the boundary to be crossed.

The deepest lesson of Chernobyl is precisely that these layers combined. A machine with the wrong sign of feedback can be operated safely *only* if the operators rigorously avoid the regimes where that sign bites—and that requires the organisation to know, document, and enforce those boundaries. All three failed together.

16.3 From INSAG-1 to INSAG-7: the evolution of understanding

The International Nuclear Safety Advisory Group (INSAG) issued two major reports.

INSAG-1 (1986), drawing heavily on the initial Soviet presentation, emphasised *operator error*: the violations of procedure—running at low power, defeating protections, depleting the ORM—were presented as the primary cause. This framing placed the weight of responsibility on the crew.

INSAG-7 (1992), after the Soviet authorities released fuller information (including the State Committee's and the Soviet scientists' own deeper analyses), substantially revised the picture. It emphasised *design deficiencies*—the positive void coefficient and especially the positive scram effect—as root causes, while still acknowledging the operational violations. In INSAG-7's assessment, the operators were drawn into a trap that the reactor's physics and the inadequate information about it had set. Its often-quoted observation is that the conditions under which the positive scram effect would be dangerous were believed never to occur, yet "appeared in almost every detail" at Chernobyl.

Remark 16.1 (Why the shift matters). The movement from "the operators broke the rules" to "the machine had flaws that made rule-breaking catastrophic, and the rules themselves were poorly justified to those who had to follow them" is the central interpretive lesson of Chernobyl. It reframes safety from a matter of operator discipline alone to a matter of *inherent design*: a reactor should not be one procedure violation away from prompt criticality, and its shutdown system should never add reactivity. Modern safety philosophy—inherent and passive safety, defence in depth, and a strong safety culture—descends directly from this reframing.

16.4 The fixes to the RBMK fleet

After the accident the surviving RBMK reactors were extensively modified to address the very feedback flaws this book has emphasised:

- Fuel enrichment was increased (from $\sim 2\%$ toward 2.4%) and additional fixed absorbers were installed, which *reduced the void coefficient* so that the power coefficient became negative across the operating range.
- The control rods were redesigned and the displacer geometry changed to *eliminate the positive scram effect*, with faster insertion and a larger number of rods kept inserted.
- A hard minimum operating reactivity margin was enforced, with automatic protection,

removing the low-ORM trap.

- Improved instrumentation and faster emergency protection were added.

These changes did not make the RBMK a Western-style reactor, but they removed the specific positive-feedback pathways that caused the accident—an implicit confirmation that those pathways were indeed the root cause.

16.5 Comparison with Western practice

It is worth stating plainly the contrast that makes Chernobyl so instructive. Western light-water reactors are designed with a *negative* void coefficient and a *negative* power coefficient, achieved by under-moderation; their shutdown systems insert only negative reactivity, quickly; and their containment buildings are designed to retain radioactivity even given core damage—a feature the RBMK lacked. The same reactivity-insertion event that destroyed Chernobyl would be self-limited in a PWR by its prompt negative feedback (the BORAX/SPERT result), and any residual release would be contained. Chernobyl is thus simultaneously a story about one flawed design and a vindication of the inherent-safety principles embodied in others.

> **Key point.** Chernobyl was a feedback failure: in its operating state the dominant reactivity feedbacks were positive, so the coupled system was unstable and the prompt excursion had no benign limit. The understanding evolved from blaming operators (INSAG-1) to recognising design flaws—the positive void coefficient and positive scram effect—as root causes (INSAG-7). The enduring lesson is that safety must be inherent in the physics, not contingent on perfect operation.

Exercises

1. Frame the accident as a feedback-loop failure using the language of Chapters 6–9.
2. Summarise the differences between INSAG-1 and INSAG-7 and why the shift matters.
3. List the design, operational, and organisational causes and argue why all three were necessary.
4. Which post-accident RBMK modifications addressed the void coefficient, and how?
5. Contrast the RBMK with a Western PWR on void coefficient, scram, and containment.

CHAPTER 17

Decay-Heat Accidents: Three Mile Island and Fukushima

Chernobyl was a reactivity-stability accident: the fission power ran away. The two other defining accidents of the nuclear age—Three Mile Island and Fukushima—were the opposite kind: the reactors shut down correctly, but the *decay heat* (Chapter 4) could not be removed. Together with Chernobyl they complete the picture of reactor thermal safety, showing that stopping fission is necessary but not sufficient.

17.1 Three Mile Island (1979)

At Three Mile Island Unit 2, a minor fault (a stuck-open pressuriser relief valve) combined with misleading instrumentation led operators to throttle back the emergency core cooling under the mistaken belief that the system was full of water. The reactor had scrammed promptly and correctly—there was no reactivity problem—but decay heat, initially several per cent of the 2700 MW thermal rating, continued unabated. With cooling reduced, the core boiled dry in part, overheated, and suffered substantial melting over the following hours. The containment building, however, held, and off-site releases were minimal.

TMI's lessons were about *human factors and decay heat*: control-room instrumentation that misled rather than informed; operator training that had not prepared the crew for the actual scenario; and the inexorability of decay heat. It transformed operator training, control-room design, and emergency procedures worldwide—and, seven years before Chernobyl, established that a reactor is a thermal hazard for days after shutdown.

17.2 Fukushima Daiichi (2011)

The Fukushima Daiichi accident is the starkest demonstration of the decay-heat imperative. A magnitude-9 earthquake caused the operating reactors to scram correctly; fission stopped as designed. But the ensuing tsunami flooded the emergency diesel generators and switchgear, causing a prolonged *station blackout*—a complete loss of AC power. With no power to drive cooling, decay heat (Chapter 4) accumulated in the cores over hours and days. The water boiled away, the fuel overheated, the zirconium cladding reacted with steam to generate hydrogen, and the hydrogen exploded, breaching containment in three units and releasing radioactivity.

Fukushima was not a stability accident and not an operator-violation accident; it was a failure to remove decay heat after a beyond-design-basis natural event disabled the active cooling systems. Its central lesson reinforced the move toward *passive* decay-heat removal (Chapter 18): cooling that needs no electrical power and no operator action for a long grace period, precisely because the active systems can be lost.

17.3 Three accidents, three lessons

The three defining accidents form a complete syllabus of reactor thermal safety:

- **Chernobyl (1986)**—a *reactivity-stability* failure: positive feedback and a flawed scram drove a prompt-critical excursion. Lesson: inherent negative feedback and a sound shutdown system are non-negotiable.
- **Three Mile Island (1979)**—a *decay-heat plus human-factors* failure: correct scram, but cooling was throttled and the core melted. Lesson: instrumentation, training, and respect for decay heat.
- **Fukushima (2011)**—a *decay-heat plus station-blackout* failure: correct scram, but a natural disaster disabled active cooling. Lesson: passive, power-independent decay-heat removal.

Stability (the subject of this book) addresses the first; passive decay-heat removal addresses the second and third. A reactor that is both inherently stable *and* passively cooled is safe against all three failure modes—the goal of modern design.

> **Key point.** Not every reactor accident is a stability accident. TMI and Fukushima shut down correctly but could not remove decay heat—one through human factors, one through a natural disaster that disabled active cooling. Thermal safety therefore demands both inherent stability (against excursions) and passive decay-heat removal (against loss of cooling).

Exercises

1. Classify TMI, Fukushima, and Chernobyl by failure mode (reactivity vs. decay heat).
2. Explain why correct scram did not prevent core damage at TMI and Fukushima.
3. What is a station blackout, and why is it central to Fukushima?
4. Argue why passive decay-heat removal addresses the Fukushima failure mode.
5. State the single lesson each of the three accidents contributes to thermal safety.

Part V

Modern Practice and Synthesis

CHAPTER 18

Inherent and Passive Safety in Modern Designs

The lesson of Chernobyl—that safety must live in the physics, not in perfect operation—has shaped reactor design ever since. This chapter surveys how modern reactors build in negative feedback and passive heat removal so that thermal stability and decay-heat safety are guaranteed by nature rather than by intervention.

18.1 Inherent vs. engineered safety

Engineered safety relies on active systems—pumps, valves, diesel generators, operator actions—to respond to upsets. *Inherent* safety relies on the physical properties of the reactor: feedback coefficients, material choices, and geometry that make dangerous states physically self-correcting or unreachable. *Passive* safety relies on natural forces—gravity, natural circulation, conduction, stored coolant—that need no power or operator action. The modern ideal combines all three with defence in depth, but assigns the foundational role to inherent and passive features, precisely because Chernobyl, TMI, and Fukushima showed that active systems and operators can fail.

18.2 Designing in negative feedback

The first principle is a **negative power coefficient across all operating states**. This is achieved through:

- **Under-moderation** in light-water reactors, guaranteeing negative moderator and void coefficients (the opposite of the RBMK).
- Strong **Doppler feedback** from fertile resonance absorbers (^{238}U, ^{240}Pu), giving a prompt negative coefficient that limits excursions.
- Avoidance of positive void pathways: no graphite-decoupled-from-coolant geometry, and shutdown systems that insert only negative reactivity.

Licensing in most countries requires demonstrating a negative power coefficient and analysing reactivity-initiated accidents to show self-limitation. The RBMK's positive void coefficient would not be licensable under such rules.

18.3 Passive decay-heat removal

Because decay heat (Chapter 4) is irreducible, modern designs provide *passive* paths to remove it without power. Examples include gravity-driven core flooding, natural-circulation cooling loops, large elevated water pools that boil off over days, and—in some advanced designs—heat conduction to the ground or air. The goal is a reactor that can survive a complete station blackout (as at Fukushima) for an extended "grace period" with no operator action and no electrical power. The AP1000, for instance, is designed to remove decay heat passively for about three days using stored water and natural circulation.

18.4 Generation III+ light-water reactors

Evolutionary designs (AP1000, EPR, and others) retain the proven negative-feedback physics of the PWR and add passive safety systems, larger water inventories, core-catchers to contain molten fuel, and digital protection. They are, in stability terms, conservative: their inherent feedback is strongly negative and their innovations are mostly in passive heat removal and containment. They embody the direct engineering response to the failure modes of TMI, Chernobyl, and Fukushima.

18.5 Generation IV and advanced reactors

More radical designs pursue inherent safety through different physics:

- **High-temperature gas reactors** (HTGR) use TRISO fuel particles that retain fission products to very high temperatures and a core whose large negative temperature coefficient and low power density allow it to shed decay heat by conduction and radiation alone—it cannot melt even with no cooling.
- **Molten-salt reactors** carry fuel dissolved in a liquid salt with a strongly negative temperature/expansion coefficient and a passive "freeze-plug" drain that empties the core into safe geometry on loss of power.
- **Sodium-cooled fast reactors** can be designed with strong negative expansion feedback; the EBR-II tests in 1986—the same year as Chernobyl—famously demonstrated survival of a complete loss of flow *without scram* purely through inherent feedback.
- **Lead-cooled and heat-pipe microreactors** pursue similar walk-away-safe goals with passive heat transport.

The unifying theme is the same principle this book has developed: make the dominant feedback strongly negative and provide a passive path for decay heat, so that the reactor is safe *by physics*.

18.6 The EBR-II demonstration: the anti-Chernobyl

It is worth pausing on a remarkable coincidence of timing. On 3 April 1986—just three weeks before Chernobyl—engineers at the Experimental Breeder Reactor II (EBR-II) in Idaho deliberately subjected the reactor to a complete *loss-of-flow-without-scram* test at full power: they cut the coolant pumps and disabled the automatic shutdown. Rather than melting, the reactor's strong *negative* reactivity feedback (from thermal expansion of the metallic fuel and structure) reduced the power as temperatures rose, and the reactor settled to a safe, low-power state on its own. The same week, two reactors on opposite sides of the world were pushed beyond their protection systems; one destroyed itself and a region, the other quietly proved that inherent negative feedback is the surest safety system of all. The contrast is the thesis of this book in a single image.

Key point. Modern reactor safety is founded on inherent negative feedback (a negative power coefficient guaranteed across all states) and passive decay-heat removal (natural forces that need no power or operators). EBR-II's 1986 loss-of-flow-without-scram test is the positive mirror image of Chernobyl: inherent negative feedback turning a would-be accident into a non-event.

Exercises

1. Distinguish inherent, passive, and engineered safety with an example of each.
2. Explain how under-moderation guarantees a negative void coefficient.
3. Describe one passive decay-heat removal strategy and the grace period it provides.
4. Summarise the EBR-II 1986 loss-of-flow-without-scram test and why it is the "anti-Chernobyl".
5. For an HTGR with TRISO fuel, explain why the core "cannot melt" even without active cooling.

CHAPTER 19

Computational Modelling of Reactor Dynamics

The theory of the preceding chapters becomes vivid when integrated numerically. This chapter describes a small, open-source point-reactor-kinetics toolkit written to accompany this book, shows the transients it produces, and uses it to make the stability principles concrete—including a numerical caricature of the Chernobyl feedback failure.

19.1 A minimal point-kinetics solver

The companion code, `reactor-kinetics`, couples validated delayed-neutron data to a stiff ODE integrator with optional temperature feedback. Its core is the six-group point kinetics of Chapter 3 together with the adiabatic feedback model of Chapters 5 and 9:

$$\frac{\mathrm{d}p}{\mathrm{d}t} = \frac{\rho(t)-\beta}{\Lambda}p + \sum_i \lambda_i \zeta_i, \qquad \frac{\mathrm{d}\zeta_i}{\mathrm{d}t} = \frac{\beta_i}{\Lambda}p - \lambda_i \zeta_i,$$
$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{P_0\, p}{C}, \qquad \rho(t) = \rho_{ext}(t) + \alpha\,(T - T_0).$$

Because $\Lambda \sim 10^{-5}\,$s while the precursor times are seconds, the system is stiff; the solver uses an implicit/adaptive method (LSODA). Delayed-neutron parameters are taken from validated ^{235}U data (Appendix A).

19.2 Verification against the inhour equation

A dynamics code is only trustworthy if it reproduces known analytical results. The toolkit's asymptotic period, measured from simulated power traces, matches the inhour equation (3.5) to better than 10^{-4} across a range of step reactivities (Figure 3.1). This agreement validates both the solver and the kinetics implementation, and it reproduces the steep collapse of the period as reactivity approaches one dollar—the quantitative warning sign of approaching prompt criticality.

19.3 A gallery of transients

Figure 3.2 collects four canonical transients computed with the toolkit, each illustrating a principle from earlier chapters:

1. **Sub-prompt step (+0.3\$).** A prompt jump (Eq. 3.4) followed by a slow rise on a stable delayed-neutron period—the controllable regime.
2. **Prompt-critical step (+1.2\$).** Runaway growth on the prompt timescale (note the logarithmic axis), the signature of crossing one dollar.
3. **Control-rod scram.** A negative ramp drives the power down, with the characteristic delayed-neutron tail as the precursors decay.
4. **Reactivity-initiated excursion with feedback.** A prompt-critical insertion limited by negative temperature feedback: the power peaks and turns over, the Nordheim–Fuchs self-limitation of Chapter 9.

The fourth panel is the crux. With negative feedback the excursion is capped and the reactor survives. The exercise below shows what happens when the feedback sign is wrong.

19.4 A numerical caricature of the Chernobyl feedback failure

The toolkit makes it easy to flip the sign of the feedback and watch stability turn into instability. Consider two runs with identical prompt-critical insertions ($\rho_0 \approx 1.05\$$):

- **Negative feedback** ($\alpha < 0$, the well-designed reactor): the fuel heats, reactivity falls, and the power excursion turns over at a finite peak, depositing a bounded energy $E_{\text{total}} \approx 2(\rho_0 - \beta)/\gamma$ (Chapter 9). The reactor self-limits.
- **Positive feedback** ($\alpha > 0$, the Chernobyl-like configuration): the rising power adds further reactivity, there is no turnover, and the power grows until the model's adiabatic assumptions break—representing, in reality, fuel fragmentation, coolant flashing, and disassembly.

This two-line change in a parameter reproduces, in caricature, the essential physics of the accident: the difference between Chernobyl and a survivable RIA is the *sign of* α. The toolkit also includes a model void/temperature feedback that becomes positive at low "power," qualitatively mimicking the RBMK's low-power instability window of Chapter 8.

Remark 19.1 (On responsible modelling). The companion code is an educational point-kinetics tool. It is deliberately *not* a high-fidelity simulation of any specific reactor: it uses lumped parameters, an adiabatic feedback, and generic data. Its value is pedagogical—to let a reader reproduce the inhour equation, watch a prompt-critical excursion, and see negative feedback cap it—not to model a real plant, which requires validated, qualified codes (e.g. coupled neutronics/thermal-hydraulics systems) far beyond a point model.

19.5 Toward higher fidelity

Real safety analysis goes well beyond point kinetics. Spatial effects (Chapter 10) demand at least nodal or full three-dimensional neutronics coupled to a multi-channel thermal-hydraulics solver; the RBMK's size made such spatial modelling essential, and core-average point parameters understated the danger. Modern coupled codes solve the time-dependent neutron-transport (or diffusion) equation on a spatial mesh, feed local powers to a thermal-hydraulic subchannel model, and return local temperatures and densities as feedback—closing the loop of Chapter 7 cell by cell. The point model of this chapter is the conceptual seed of those codes, and the place to build intuition before trusting their output.

Key point. A small point-kinetics code reproduces the inhour equation to 10^{-4}, displays the canonical transients, and—by flipping the sign of the feedback coefficient—turns a self-limiting excursion into a Chernobyl-like runaway. The sign of the feedback is, numerically as well as physically, the whole story.

Exercises

1. Why must a point-kinetics solver use a stiff integration method?
2. Describe how to validate a dynamics code against the inhour equation.
3. Explain what changes in the model output when the feedback coefficient sign is flipped.
4. What are the limitations of a lumped point-kinetics model for analysing a real, large reactor?
5. Outline what a higher-fidelity coupled neutronics/thermal-hydraulics code adds over a point model.

CHAPTER 20

Synthesis and Conclusions

We have travelled from the neutron balance to the ruins of Chernobyl Unit 4. This closing chapter draws the threads together into a small number of durable principles.

20.1 The one-sentence theory

If this book could be compressed to a sentence, it is this:

> *A reactor is thermally stable if and only if a rise in power removes reactivity—through temperature and density feedback—faster and more surely than the rise in power can add it; and this must hold promptly, and at every power level and core state the reactor can reach.*

Every chapter is an elaboration. Point kinetics (Chapter 3) sets the timescale on which "faster" must be measured and identifies one dollar as the cliff edge. Thermal transport (Chapter 5) sets how quickly temperature follows power, and which feedbacks are prompt. The feedback coefficients (Chapter 6) supply the signs and magnitudes. Stability theory (Part III) turns "faster and more surely" into eigenvalues and Nyquist loci. And Chernobyl (Part IV) shows what the violation looks like.

20.2 Five principles of thermally stable design

1. A negative power coefficient, always. The aggregate of all temperature/void feedbacks must remove reactivity as power rises, at every operating point. This single property makes a reactor self-regulating. The RBMK's low-power positive coefficient was the root flaw.

2. Prompt negative feedback to cap excursions. Doppler feedback, being prompt and reliably negative, is the only defence against a prompt-critical excursion. A reactor must have enough of it that the net prompt feedback is negative even with the void term included.

3. A shutdown system that only ever helps. The emergency shutdown must insert negative reactivity, quickly, under all conditions. A scram that can add reactivity—as the RBMK's graphite-tipped rods did—is a contradiction in terms and was decisive at Chernobyl.

4. Respect the slow instabilities and the memory. Xenon poisoning and spatial oscillations give the reactor an hours-long memory and can make a large core locally unstable even when globally benign. Operation must account for history and spatial shape, and negative feedback must be strong enough to damp spatial modes.

5. Plan for the heat that will not switch off. Decay heat is several per cent of full power for a long time after shutdown. A passive, reliable path to the ultimate heat sink must always exist—the lesson of TMI and Fukushima as much as of Chernobyl.

20.3 The human and organisational dimension

The physics is necessary but not sufficient. Chernobyl's deepest lesson, captured in the shift from INSAG-1 to INSAG-7, is that a machine with marginal or wrong-sign feedback places an impossible burden on its operators: it asks them never to enter states whose danger they were never properly told about. Safety culture—honest documentation, conservative procedures, the freedom to halt a test, and the humility to treat margins as physics rather than paperwork—is itself a part of the safety system. The best reactors reduce this burden by being inherently safe; the best organisations carry what remains of it conscientiously.

20.4 The two reactors of April 1986

We end where the contrast is sharpest. In April 1986, two reactors were deliberately pushed past their protection systems. At EBR-II in Idaho, engineers cut the coolant flow at full power without scram, and the reactor's strong negative feedback quietly brought it to a safe state—a planned demonstration of inherent safety. Three weeks later at Chernobyl, a reactor with a positive void coefficient and a flawed scram was driven into its least stable regime, and its physics tore it apart. Same era, same act of pushing a reactor past its protections, opposite outcomes—decided by the *sign of the feedback*. That sign is the whole of thermal stability, and it is the legacy that the victims of Chernobyl bequeathed to every reactor designed since.

20.5 Closing

Thermal stability is not an exotic corner of reactor physics; it is the property that makes nuclear power possible. Understood, it yields machines that ride out disturbances and shut themselves down. Ignored or got wrong, it yields Chernobyl. The mathematics in this book is, in the end, a long and careful way of saying something simple: build reactors whose own nature opposes their excursions, and never operate a reactor in a state where that nature is reversed.

Exercises

1. State the one-sentence theory of thermal stability in your own words.
2. List the five principles of thermally stable design and give the Chernobyl counter-example for each.
3. Argue why safety culture is part of the safety system, citing the INSAG-1 to INSAG-7 shift.
4. Compare the EBR-II and Chernobyl events of April 1986 and state what single quantity decided their outcomes.
5. In what sense is thermal stability "the property that makes nuclear power possible"?

Appendices

APPENDIX A

Delayed-Neutron and Kinetics Data

The transients and figures in this book use the six-group delayed-neutron data for thermal fission of ^{235}U listed in Table A.1. These values are representative of the validated parameters distributed with the PyRK package and are consistent with standard compilations (Keepin; ENDF-derived sets). The total delayed fraction is $\beta = \sum_i \beta_i$, and a representative prompt neutron generation time for a thermal system is $\Lambda \approx 1.1 \times 10^{-5}$ s.

Table A.1: Six-group delayed-neutron data for thermal fission of ^{235}U (representative values).

Group i	β_i	fraction β_i/β	λ_i [s^{-1}]
1	0.000247	0.033	0.0124
2	0.001385	0.219	0.0305
3	0.001222	0.196	0.111
4	0.002645	0.395	0.301
5	0.000832	0.115	1.14
6	0.000169	0.042	3.01
Total	$\beta \approx 0.0065$	1.000	—

The exact group fractions and decay constants vary slightly among evaluations and with the fissioning isotope and spectrum. For ^{239}Pu the total delayed fraction is smaller ($\beta \approx 0.0021$), which makes plutonium-bearing cores reach prompt criticality at a smaller absolute reactivity—a consideration for fast reactors and high-burnup fuel.

Table A.2: Representative kinetics parameters used in the worked examples.

Quantity	Symbol	Value
Total delayed fraction	β	6.5×10^{-3}
Prompt generation time (thermal)	Λ	$\sim 1.1 \times 10^{-5}$ s
One dollar	$\$ = \beta$	650 pcm
Mean delayed-neutron lifetime	$\sum \beta_i/(\beta\lambda_i)$	~ 13 s

The wide separation between Λ (10^{-5} s) and the mean delayed-neutron lifetime (~ 13 s) is the numerical statement of why delayed neutrons make reactors controllable, and why crossing $\rho = \beta$ removes that protection.

APPENDIX B

Selected Derivations

B.1 The inhour equation from point kinetics

Starting from the point kinetics (3.2) with constant ρ, seek solutions $n = n_0 e^{st}$ and $C_i = C_{i0} e^{st}$. The precursor equation gives

$$s\,C_{i0} = \frac{\beta_i}{\Lambda} n_0 - \lambda_i C_{i0} \;\Rightarrow\; C_{i0} = \frac{\beta_i\, n_0}{\Lambda(s+\lambda_i)}. \tag{B.1}$$

Substituting into the neutron equation,

$$s\,n_0 = \frac{\rho-\beta}{\Lambda} n_0 + \sum_i \lambda_i \frac{\beta_i n_0}{\Lambda(s+\lambda_i)}. \tag{B.2}$$

Divide by n_0/Λ and rearrange, using $\lambda_i/(s+\lambda_i) = 1 - s/(s+\lambda_i)$:

$$\Lambda s = \rho - \beta + \sum_i \beta_i \Big(1 - \frac{s}{s+\lambda_i}\Big) = \rho - \sum_i \frac{\beta_i s}{s+\lambda_i}, \tag{B.3}$$

which gives the inhour equation (3.5), $\rho = \Lambda s + \sum_i \beta_i s/(s+\lambda_i)$. The function on the right has simple poles at $s = -\lambda_i$ and is monotonic between consecutive poles, guaranteeing exactly one real root in each of the intervals separated by the poles—hence $G+1$ real roots for G groups, with the dominant root carrying the sign of ρ.

B.2 The prompt-jump approximation

For a step $\rho < \beta$ at $t = 0$, immediately after the step the precursor concentrations retain their pre-step equilibrium values $C_{i0} = \beta_i n_0/(\lambda_i \Lambda)$, so $\sum_i \lambda_i C_{i0} = \beta n_0/\Lambda$. Treating the prompt dynamics as instantaneous, set $\mathrm{d}n/\mathrm{d}t \approx 0$:

$$0 \approx \frac{\rho-\beta}{\Lambda} n(0^+) + \frac{\beta}{\Lambda} n(0^-) \;\Rightarrow\; n(0^+) = \frac{\beta}{\beta-\rho}\, n(0^-), \tag{B.4}$$

the prompt-jump factor (3.4). The approximation holds when the prompt timescale $\Lambda/(\beta-\rho)$ is much shorter than the precursor timescales $1/\lambda_i$, i.e. away from prompt criticality.

B.3 The Nordheim–Fuchs energy release

From Chapter 9, with prompt kinetics and adiabatic feedback $\rho = \rho_0 - \gamma E$, $\mathrm{d}E/\mathrm{d}t = p$, the power as a function of energy is

$$p(E) = p_0 + \frac{1}{\Lambda}\Big[(\rho_0-\beta)E - \tfrac{1}{2}\gamma E^2\Big]. \tag{B.5}$$

The peak occurs at $\mathrm{d}p/\mathrm{d}E = 0$, i.e. $\gamma E_{\text{peak}} = \rho_0 - \beta$. Neglecting p_0, the excursion ends ($p \to 0$) when the bracket returns to zero, at $E_{\text{total}} = 2(\rho_0 - \beta)/\gamma$, twice the peak energy by the symmetry of the parabola. The peak power follows by substituting E_{peak}: $p_{\max} = (\rho_0 - \beta)^2/(2\gamma\Lambda)$, and the adiabatic temperature rise is $\Delta T = E_{\text{total}}/C$.

APPENDIX C

Detailed Chernobyl Timeline

The following approximate timeline summarises the principal events of 25–26 April 1986 (local time). Times and some power values vary slightly among sources; Table C.1 follows the broad consensus of the World Nuclear Association sequence and INSAG-7.

Table C.1: Approximate sequence of events, Chernobyl Unit 4.

Time (approx.)	**Event**
25 Apr, 01:06	Planned power reduction for the turbine-coastdown test begins.
25 Apr, 03:47	Power reduced to about half ($\sim$1600 MW_{th}).
25 Apr, 14:00	ECCS isolated for the test; grid dispatcher asks the unit to keep generating, delaying the test for hours.
26 Apr, 00:05	Power reduction resumes toward the test level.
26 Apr, 00:28	Power falls far below plan, to roughly 30 MW_{th}; severe xenon poisoning develops.
26 Apr, 00:43	Turbine-trip protection blocked to allow repeating the test.
26 Apr, 01:00	Power coaxed up to $\sim$200 MW_{th} by withdrawing many control rods; operating reactivity margin far below the minimum.
26 Apr, 01:03–01:07	Additional main coolant pumps started, raising flow and suppressing voids (pushing pumps toward cavitation).
26 Apr, 01:23:04	Turbine steam valves closed; coastdown test begins; flow decreases, boiling and voids increase, power begins to rise.
26 Apr, 01:23:40	AZ-5 emergency scram pressed; rods begin to insert.
26 Apr, 01:23:43–47	Positive scram effect and positive void feedback drive the reactor prompt-critical; power surges to tens of GW_{th}.
26 Apr, $\sim$01:23:45	First (steam) explosion blows off the upper shield; a second explosion follows seconds later; core destroyed, graphite ignites.
26 Apr onward	Graphite fire burns for days; large radioactive release continues for about ten days before subsiding.

The compressed final interval—from the first sign of power rise to the explosions, a matter of seconds—is the prompt-kinetics timescale of Chapter 3 made manifest. Everything that mattered physically happened faster than any human or mechanical response.

APPENDIX D

Worked Numerical Examples

This appendix collects fully worked numerical examples that tie the theory to concrete numbers. Throughout, $\beta = 0.0065$, $\Lambda = 1.1 \times 10^{-5}$ s, and the six-group ^{235}U data of Appendix A are used unless stated.

Example E.1: Reactivity conversions

A reactor has $k_{\text{eff}} = 1.0013$. The reactivity is $\rho = (k_{\text{eff}} - 1)/k_{\text{eff}} = 0.0013/1.0013 \approx 1.30 \times 10^{-3} = 130\,\text{pcm}$. In dollars, $\$ = \rho/\beta = 0.0013/0.0065 = 0.20\,\$$. This is a modest, delayed-critical insertion—well below the one-dollar prompt-critical threshold.

Example E.2: Prompt jump

For a step insertion of $\rho = 0.20\,\$ = 0.0013$, the prompt-jump factor is $\beta/(\beta - \rho) = 0.0065/(0.0065 - 0.0013) = 0.0065/0.0052 = 1.25$. Power jumps essentially instantly to $1.25\times$ its initial value, then rises slowly on the delayed-neutron period.

Example E.3: Stable period from the inhour equation

For the same $\rho = 0.0013$, a one-delayed-group estimate with effective $\bar{\lambda} \approx 0.08\,\text{s}^{-1}$ gives a dominant root from $\rho \approx \beta s/(s + \bar{\lambda})$, i.e. $s \approx \bar{\lambda}\,\rho/(\beta - \rho) = 0.08 \times 0.0013/0.0052 \approx 0.02\,\text{s}^{-1}$, so the period $T = 1/s \approx 50\,\text{s}$. The full six-group inhour solution gives a similar tens-of-seconds period (cf. Figure 3.1), confirming a slow, controllable transient.

Example E.4: Prompt-critical excursion

For $\rho = 1.5\,\$ = 0.00975$, the prompt reactivity excess is $\rho - \beta = 0.00325$. The prompt growth rate is $(\rho - \beta)/\Lambda = 0.00325/1.1 \times 10^{-5} \approx 295\,\text{s}^{-1}$, an e-folding time of $\sim 3.4\,\text{ms}$. In $20\,\text{ms}$ the power grows by $e^{295 \times 0.02} = e^{5.9} \approx 365$. This is the regime in which only prompt feedback—not control—can intervene.

Example E.5: Nordheim–Fuchs energy release

Take $\rho_0 = 1.2\,\$ = 0.0078$, so $\rho_0 - \beta = 0.0013$. Suppose the feedback parameter is $\gamma = |\alpha|/C$ with $|\alpha| = 3 \times 10^{-5}\,\mathrm{K}^{-1}$ and $C = 5 \times 10^4\,\mathrm{J/K}$, giving $\gamma = 6 \times 10^{-10}\,\mathrm{J}^{-1}$. The energy at peak is $E_{\mathrm{peak}} = (\rho_0 - \beta)/\gamma = 0.0013/6 \times 10^{-10} \approx 2.2 \times 10^6\,\mathrm{J}$, and the total prompt-spike energy is about $4.3 \times 10^6\,\mathrm{J}$. The adiabatic fuel temperature rise is $\Delta T = E_{\mathrm{total}}/C \approx 87\,\mathrm{K}$—a survivable spike, illustrating self-limitation by negative feedback. Had α been positive, no such cap would exist.

Example E.6: Decay-heat heat-up

A $3000\,\mathrm{MW_{th}}$ core loses cooling just after shutdown with 6% decay heat, i.e. $q = 1.8 \times 10^8\,\mathrm{W}$. With core thermal mass $C = 1.0 \times 10^7\,\mathrm{J/K}$, the adiabatic heat-up rate is $\dot{T} = q/C = 18\,\mathrm{K/s}$. To rise $2000\,\mathrm{K}$ toward fuel melting takes about $110\,\mathrm{s}$—under two minutes—quantifying the absolute need for assured decay-heat removal.

Example E.7: Void feedback loop gain (illustrative)

Suppose a power increase of 10% raises the core-average void fraction by 0.02, and the void coefficient is $\alpha_v = +1.5\,\$$ per unit void fraction (an RBMK-like positive value at low power). The feedback reactivity is $\Delta\rho = +0.02 \times 1.5 = +0.03\,\$$. Positive feedback of this sign and size, unopposed by sufficient prompt Doppler, drives the power further up—the unstable loop. The same calculation with $\alpha_v < 0$ (a PWR) yields negative feedback and a self-correcting response.

APPENDIX E

Mathematical Methods

This appendix collects the linear-systems and stability results invoked in the main text, with statements and short proofs, so the book is self-contained.

E.1 Linear systems and the matrix exponential

For $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, $\mathbf{x}(0) = \mathbf{x}_0$, the unique solution is $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0$, where $e^{\mathbf{A}t} = \sum_{k\geq 0}(\mathbf{A}t)^k/k!$ converges for all t.

Theorem E.1 (Spectral mapping)**.** *If* $\mathbf{A}$ *has eigenvalues* $\{s_j\}$ *then* $e^{\mathbf{A}t}$ *has eigenvalues* $\{e^{s_j t}\}$ *with the same eigenvectors. Consequently* $\|\mathbf{x}(t)\| \to 0$ *for every* $\mathbf{x}_0$ *iff* $\Re s_j < 0$ *for all* j*.*

Proof. If $\mathbf{A}\mathbf{v} = s\mathbf{v}$ then $e^{\mathbf{A}t}\mathbf{v} = \sum_k (st)^k/k!\,\mathbf{v} = e^{st}\mathbf{v}$. Decomposing $\mathbf{x}_0$ in a Jordan basis, each mode evolves as $t^m e^{s_j t}$; this tends to 0 for all m iff $\Re s_j < 0$. □

E.2 Lyapunov's theorem for linear systems

Theorem E.2 (Lyapunov)**.** $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ *is asymptotically stable iff for some (every) symmetric positive-definite* $\mathbf{Q} \succ 0$ *the Lyapunov equation* $\mathbf{A}^\top\mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}$ *has a unique symmetric solution* $\mathbf{P} \succ 0$*. Then* $V(\mathbf{x}) = \mathbf{x}^\top\mathbf{P}\mathbf{x}$ *satisfies* $\dot{V} = -\mathbf{x}^\top\mathbf{Q}\mathbf{x} < 0$*.*

Proof. If $\mathbf{A}$ is Hurwitz, $\mathbf{P} = \int_0^\infty e^{\mathbf{A}^\top t}\mathbf{Q}\,e^{\mathbf{A}t}\,dt$ converges, is positive definite, and satisfies the Lyapunov equation (differentiate under the integral). Conversely, if such $\mathbf{P} \succ 0$ exists, then $V > 0$ and $\dot{V} < 0$ force $\mathbf{x}(t) \to 0$, so $\mathbf{A}$ is Hurwitz. □

E.3 The Routh–Hurwitz criterion

Theorem E.3 (Routh–Hurwitz)**.** *The real polynomial* $s^n + a_1 s^{n-1} + \cdots + a_n$ *has all roots with negative real part iff all leading principal minors of the Hurwitz matrix are positive. In particular:*

- $n = 2$*: stable iff* $a_1 > 0$ *and* $a_2 > 0$*.*
- $n = 3$*: stable iff* $a_1 > 0$*,* $a_3 > 0$*, and* $a_1 a_2 > a_3$*.*
- $n = 4$*: stable iff* $a_1, a_2, a_3, a_4 > 0$ *and* $a_1 a_2 a_3 > a_3^2 + a_1^2 a_4$*.*

Remark on proof. The criterion follows from the Hermite–Biehler theorem on the interlacing of the roots of the even and odd parts of the polynomial along the imaginary axis; equivalently from the Routh array, whose first-column sign changes count right-half-plane roots. The low-order cases are obtained by direct evaluation of the Hurwitz determinants. □

E.4 The argument principle and Nyquist

Theorem E.4 (Argument principle)**.** *Let f be meromorphic inside and on a positively oriented simple closed contour Γ, with no zeros or poles on Γ. Then $\frac{1}{2\pi i}\oint_\Gamma \frac{f'(s)}{f(s)}\,\mathrm{d}s = Z - P$, the number of zeros minus poles of f inside Γ (with multiplicity).*

Applying this to $f = 1 + L(s)$ on the Nyquist contour yields Theorem 8.4: the encirclements of -1 by L equal $Z - P$.

E.5 Metzler matrices and positivity

Theorem E.5 (Positivity of Metzler flows)**.** *A real matrix $\mathbf{A}$ is* Metzler *(all off-diagonal entries ≥ 0) iff $e^{\mathbf{A}t} \geq 0$ entrywise for all $t \geq 0$. A Metzler $\mathbf{A}$ has a real dominant eigenvalue (its spectral abscissa) with a nonnegative eigenvector (Perron–Frobenius for Metzler matrices).*

Proof. Pick c with $\mathbf{B} = \mathbf{A} + c\mathbf{I} \geq 0$. Then $e^{\mathbf{A}t} = e^{-ct}e^{\mathbf{B}t}$ and $e^{\mathbf{B}t} = \sum_k (\mathbf{B}t)^k/k! \geq 0$. Conversely, if $e^{\mathbf{A}t} \geq 0$ for all small t, then $\mathbf{A} = \lim_{t\to 0^+}(e^{\mathbf{A}t} - \mathbf{I})/t$ has nonnegative off-diagonal entries. The Perron–Frobenius statement applies to $\mathbf{B} \geq 0$ and transfers to $\mathbf{A}$ by the shift. □

This theorem underlies Proposition 3.5: the point-kinetics generator is Metzler, so power and precursor populations stay nonnegative.

E.6 Gershgorin disc theorem

Theorem E.6 (Gershgorin)**.** *Every eigenvalue of $\mathbf{A} = (a_{ij}) \in \mathbb{C}^{n\times n}$ lies in at least one disc $\{z : |z - a_{ii}| \leq \sum_{j\neq i} |a_{ij}|\}$.*

Proof. For eigenpair $(s, \mathbf{v})$ pick i with $|v_i| = \max_k |v_k| > 0$. The ith row of $\mathbf{Av} = s\mathbf{v}$ gives $(s - a_{ii})v_i = \sum_{j\neq i} a_{ij}v_j$; dividing by v_i and bounding $|v_j/v_i| \leq 1$ yields the disc inequality. □

Gershgorin discs give quick, rigorous bounds on the kinetic and thermal eigenvalues without solving the characteristic polynomial—useful for verifying, for instance, that all delayed-precursor eigenvalues remain near their poles $-\lambda_i$ under weak feedback.

APPENDIX F

Glossary

Criticality
The state $k_{\text{eff}} = 1$ in which neutron production balances loss and the chain reaction is self-sustaining at constant power.

Delayed neutrons
The small fraction (β) of fission neutrons emitted seconds to minutes after fission by the decay of precursor fission products; they set the controllable timescale of a reactor.

Doppler coefficient
The (prompt, negative) change in reactivity per unit fuel temperature, arising from broadening of ^{238}U resonances.

Dollar
A unit of reactivity equal to β; one dollar marks prompt criticality.

Decay heat
Thermal power from radioactive decay of fission products, several per cent of full power just after shutdown, declining slowly over days.

Inhour equation
The relation between a constant reactivity and the reactor period, derived from point kinetics.

Operating reactivity margin (ORM)
The RBMK's shutdown reserve, expressed as an equivalent number of inserted control rods; a minimum was mandated and was grossly violated at Chernobyl.

Point kinetics
The reduction of reactor dynamics to ordinary differential equations for the spatially-averaged power and precursor concentrations.

Positive scram effect
The RBMK phenomenon in which beginning to insert the graphite-tipped control rods added positive reactivity in the lower core.

Prompt criticality
The state $\rho \geq \beta$ in which prompt neutrons alone sustain growth; the response collapses to the prompt timescale.

Reactivity
The fractional departure from criticality, $\rho = (k_{\text{eff}} - 1)/k_{\text{eff}}$.

Void coefficient
The change in reactivity per unit coolant void (steam) fraction; negative in under-moderated water reactors, strongly positive in the RBMK.

Xenon-135
A fission product with an enormous neutron-absorption cross-section whose hours-long dynamics cause post-shutdown poisoning and spatial oscillations.

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